



Enhancing Loan Default Prediction Using Topological Data Analysis

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Abstract

Accurate prediction of loan defaults is a crucial challenge for financial institutions, and while traditional machine learning models have made significant contributions, they often fail to capture the complex, non-linear relationships inherent in financial data. This research explores the integration of Topological Data Analysis (TDA) with machine learning models to enhance the prediction of loan defaults. By incorporating persistent homology and barcode statistics, TDA-derived features are added to traditional machine learning algorithms such as K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, Support Vector Machines (SVM), and XGBoost.

The study evaluates the effectiveness of TDA-enhanced models through four experiments, comparing them to traditional models using standard evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. The results demonstrate significant improvements in model performance, particularly for XGBoost and SVM, with TDA-derived features enhancing predictive accuracy and interpretability. Hyperparameter tuning was found to further improve model stability and performance. The integration of landmark sampling enabled efficient handling of large datasets, balancing computational costs and model accuracy.

Despite these promising results, the study acknowledges several limitations, including the reliance on static publicly available datasets, the computational cost of TDA, and the challenges associated with interpretability and hyperparameter tuning. The study also suggests avenues for future research, such as expanding the scope to more diverse datasets, exploring additional machine learning models, optimizing TDA computation for real-time applications, and improving the interpretability of TDA-derived features.

This research contributes to the growing body of literature on TDA in machine learning, highlighting its potential to enhance predictive models in the financial sector. The findings suggest that TDA can be a valuable tool for improving loan default prediction, providing deeper insights into financial data and offering a more interpretable approach for high-stakes decision-making processes in finance.

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Chapter 1

Introduction

1.1 Background of the Study

Loan default prediction is a critical area of focus for financial institutions, as accurately forecasting which loans are likely to default enables these organizations to manage risk effectively, minimize financial losses, and make informed lending decisions. Traditionally, loan default prediction has relied on statistical and machine learning models that analyze structured, often tabular, financial data, identifying patterns that suggest an increased risk of default. Commonly used models for this purpose include logistic regression, decision trees, and random forests, which are effective to an extent but can face limitations when dealing with highly complex or non-linear data patterns [Anderson, 2007].

In recent years, advancements in data science have allowed for exploring innovative techniques beyond traditional models to improve prediction accuracy. One such advancement is Topological Data Analysis (TDA), a branch of data science that uses mathematical topology to study the shape and structure of data [Carlsson, 2009]. Unlike classical machine learning, which relies heavily on features derived from linear relationships, TDA captures the intrinsic geometric and topological properties within data. By examining the "shape" of data, TDA has the potential to reveal underlying patterns and relationships that traditional methods may miss. This is particularly promising in financial datasets, where data points often exhibit complex relationships and irregular structures.

Given these capabilities, TDA is a compelling approach for enhancing loan default prediction models. The primary goal of this research is to integrate TDA, particularly through techniques like persistent homology, into standard machine learning frameworks to evaluate if it enhances predictive accuracy and robustness in loan default modeling. This study utilizes two key datasets: the **Default of Credit Card Clients** dataset and the **Statlog German Credit** dataset, each providing unique insights into borrower behavior. The Default of Credit Card Clients dataset, comprising 30,000 instances, provides a large, complex dataset to test model scalability. Meanwhile, the Statlog German Credit dataset, with 1,000 instances, allows for the application of TDA on a smaller scale, supporting a focused examination of TDA's performance across dataset sizes and structures.

The integration of TDA into loan default prediction models aligns with a broader industry goal: developing predictive tools that are not only accurate but also provide transparency and deeper interpretability. By leveraging TDA's capacity to reveal patterns embedded in high-dimensional data, this research aims to support financial institutions in developing reliable risk assessment models. Furthermore, this study contributes to the academic discourse on applying TDA to real-world problems, expanding its application from theoretical explorations to practical, high-impact areas such as financial risk management.

1.2 Problem Statement

Despite the advancements in machine learning, conventional methods for loan default prediction face several challenges. These models often struggle with complex data structures, such as non-linear relationships and high-dimensional variables, which are common in financial datasets [Anderson, 2007]. As a result, they may fail to capture all critical patterns, leading to models with limited predictive power and insufficient risk differentiation. This is particularly problematic for financial institutions where accurate predictions directly impact lending strategies and financial stability.

Topological Data Analysis presents a promising alternative, as it captures data’s underlying topology, potentially unveiling complex structures that traditional models overlook. However, integrating TDA within a predictive modeling framework remains underexplored in financial applications. This thesis addresses this gap by investigating whether TDA can improve the performance of machine learning models in predicting loan defaults, thereby offering a more nuanced approach to risk assessment.

1.3 Research Objectives

This study has three primary objectives:

1. **To assess the predictive capability of traditional machine learning models** (e.g., logistic regression, decision trees) on the selected datasets and establish a baseline for prediction accuracy.
2. **To evaluate the effectiveness of TDA techniques, particularly persistent homology**, in extracting topological features from the financial datasets.
3. **To integrate TDA-derived features into traditional machine learning models** and compare their predictive performance against models without TDA enhancements, using metrics such as accuracy, precision, recall, and f1-score.

1.4 Research Questions

The research addresses the following questions:

- To what extent can TDA enhance the predictive accuracy of machine learning models in loan default prediction?
- How do topological features derived from persistent homology contribute to model performance?
- What are the comparative strengths and weaknesses of traditional versus TDA-enhanced models in loan default prediction?

1.5 Significance of the Study

This research is significant for both academia and industry. For academic research, it explores TDA’s potential in a novel context—loan default prediction—expanding the application of TDA beyond theoretical studies to practical, data-driven insights in financial analytics. From an industry perspective, incorporating TDA in predictive modeling has the potential to improve loan default prediction, ultimately enabling financial institutions to make more informed lending decisions, manage risk effectively, and minimize financial losses. This study thus supports the development of advanced data analysis tools in financial risk management, contributing to more transparent and interpretable predictive models.

1.6 Scope and Delimitations

This study focuses on applying TDA to loan default prediction using two specific datasets: the Default of Credit Card Clients dataset and the Statlog German Credit dataset. The project is limited to assessing TDA’s impact on predictive accuracy within the financial sector, and findings may not directly extend to other applications of TDA without additional validation. Moreover, only select machine learning models are utilized as a benchmark, which limits the scope to a comparison between baseline and TDA-enhanced models within these datasets.

Chapter 2

Literature Review

2.1 Overview of Loan Default Prediction

Predicting loan default risk has long been a priority for financial institutions, as it directly impacts lending policies, financial planning, and overall profitability. The ability to accurately classify loans as “high-risk” or “low-risk” helps financial institutions mitigate losses and set appropriate lending criteria. Consequently, a substantial body of research has developed around loan default prediction, typically employing binary classification models that assess whether a loan will default or remain in good standing based on specific borrower characteristics and financial metrics [Brown and Mues, 2012].

Historically, **statistical models** such as logistic regression have been widely used for loan default prediction. Logistic regression offers simplicity and interpretability, which makes it popular for binary classification problems in finance. However, logistic regression has limitations, particularly when dealing with high-dimensional data or non-linear relationships within datasets [Lee and Chen, 2005]. As datasets have grown in complexity, with numerous interdependent financial variables, the capacity of traditional statistical models to capture nuanced relationships has diminished, leading researchers to explore more sophisticated methods.

To address these challenges, **machine learning models** like decision trees, random forests, and support vector machines (SVMs) have gained popularity in loan default prediction. Decision trees and random forests, for example, are particularly valuable due to their ability to handle non-linear relationships, flexibility with large datasets, and relatively high predictive accuracy [Thomas, 2000]. Random forests, an ensemble learning method, construct multiple decision trees and aggregate their predictions, resulting in enhanced accuracy and robustness compared to single-tree models [Breiman, 2001].

Another advancement in default prediction is gradient boosting. Techniques like Gradient Boosted Decision Trees (GBDT) and XGBoost have shown promising results, particularly in handling large, structured datasets. These models build multiple weak learners in a sequential manner, adjusting each model to correct the errors of its predecessors, thereby improving accuracy over traditional models [Friedman, 2001]. However, while these models demonstrate improved predictive accuracy, they often lack transparency, as their complexity makes it difficult to interpret individual predictions which is a concern in financial contexts where transparency is crucial for regulatory compliance [Rudin, 2019].

Despite these advancements, traditional and even machine-learning models face limitations. Primarily, these models are constrained by their dependence on explicit feature engineering and a limited capacity to capture **complex, non-linear topologies** within financial data. Financial datasets often contain irregular, multidimensional relationships that are not fully represented through linear transformations or standard machine-learning techniques [Brown and Mues, 2012]. As a result, recent research has focused on identifying methodologies that can capture the intricate “shape” of data and reveal patterns that are otherwise challenging to detect through traditional modeling approaches. This recognition of non-linear, topological structures within financial data has set the stage for the exploration of **Topological Data Analysis (TDA)** as a means to address these gaps.

2.2 Machine Learning Approaches in Financial Prediction

The application of machine learning techniques in financial prediction, particularly loan default prediction, has grown rapidly over the last two decades. As financial datasets have become more complex,

capturing non-linear patterns and interactions has become essential. Advanced machine learning models, including neural networks, ensemble methods, and hybrid techniques, have provided alternatives to traditional statistical models, achieving high predictive accuracy and robustness in financial settings.

Neural networks are one of the most influential advancements in machine learning for financial prediction. Particularly in loan default prediction, deep learning models have demonstrated their ability to capture complex, non-linear relationships through multiple hidden layers that simulate human learning. These models effectively model high-dimensional and interdependent financial variables, making them useful for extracting nuanced patterns within complex datasets [Hayashi, 2022]. However, neural networks are often considered "black-box" models due to their lack of interpretability which is a significant drawback in regulated financial sectors that require transparent decision-making processes [Brown and Mues, 2012].

Ensemble learning techniques, such as bagging, boosting, and stacking, have also gained traction for financial prediction due to their capacity to improve model accuracy and generalizability. One popular ensemble method is **Gradient Boosted Decision Trees (GBDT)**, which have been shown to perform well in financial risk prediction, offering high accuracy and resilience against overfitting [Chen and Guestrin, 2016]. However, similar to neural networks, ensemble methods often suffer from interpretability issues. Models like XGBoost involve complex transformations and feature interactions that can be difficult to interpret, making them less transparent for risk assessment purposes [Akinjole et al., 2024].

Another widely applied ensemble method is **Random Forests**. Random forests combine multiple decision trees to improve prediction accuracy, handling both linear and non-linear relationships effectively. Their robustness against overfitting and flexibility in dealing with missing data and diverse feature types have made random forests particularly suitable for financial prediction tasks [Breiman, 2001]. Compared to GBDT, random forests offer a degree of interpretability, as individual trees can be examined; however, the aggregated decision-making process is still challenging to fully interpret [Thomas, 2000].

Hybrid models that combine multiple machine learning techniques are increasingly being explored to enhance predictive accuracy. For instance, models that integrate neural networks with traditional statistical methods or ensemble approaches have shown promise in loan default prediction. A two-stage hybrid model, for example, may apply decision trees to pre-select key predictive features before feeding these features into a neural network for classification, optimizing both accuracy and feature importance analysis [Li et al., 2021]. These hybrid approaches aim to leverage the strengths of multiple methods to produce a more accurate and interpretable model.

Despite these advancements, machine learning models face challenges in handling the complex, non-linear structures and high-dimensionality typical in financial datasets. Many of these models rely heavily on explicit feature engineering and linear transformations, which may limit their ability to capture intricate data patterns. Recent research has begun exploring more novel methodologies, such as Topological Data Analysis (TDA), which seeks to analyze the "shape" of data and extract meaningful topological features that can complement machine learning models. Integrating TDA with machine learning offers a promising pathway to address the limitations in traditional models by allowing a deeper exploration of the complex data structures often found in financial datasets.

2.3 Introduction to Topological Data Analysis (TDA)

Topological Data Analysis (TDA) is an emerging field within data science that leverages concepts from algebraic topology to analyze the "shape" or structure of data. Unlike traditional statistical methods that primarily focus on linear relationships, TDA examines complex geometric and topological structures in high-dimensional datasets, offering a unique perspective on data patterns. TDA has shown promise in fields requiring the analysis of intricate and high-dimensional data, including genomics, neuroscience, and, more recently, financial analytics [Carlsson, 2009].

2.3.1 Theoretical Foundations of TDA

TDA is grounded in the mathematical study of topology, which is concerned with properties that are preserved under continuous transformations, such as stretching or bending, but not tearing or gluing. In the context of data analysis, these properties can reveal critical insights into data structure, such as clusters, holes, and connected components, that are robust to noise and other distortions [Zomorodian, 2005]. One of the key aspects of TDA is its ability to provide a multi-scale view of data: by examining data at

different levels of granularity, TDA captures both local and global features, making it particularly useful for high-dimensional data.

Persistent Homology is one of the primary techniques in TDA and is widely used to identify topological features within data. Persistent homology captures the birth and death of topological features (e.g., connected components, loops, and voids) as the scale of observation changes. This process involves constructing a sequence of simplicial complexes—a combinatorial representation of data that generalizes the concept of points, edges, and higher-dimensional simplices [Edelsbrunner and Harer, 2010]. By tracking the persistence of these features across different scales, persistent homology provides insights into which topological structures are significant and which are likely to be noise. This persistence information is often summarized in visual representations known as **persistence diagrams** or **barcodes**, which graphically display the lifespan of each topological feature [Ghrist, 2008].

For example, in the case of loan default prediction, persistent homology can be used to analyze the dataset’s geometric structure, revealing clusters or other topological features that may correspond to groups of borrowers with similar risk profiles. These topological features can provide additional insights that are not easily accessible through traditional machine learning models, potentially improving the model’s predictive performance.

The Mapper Algorithm is another popular method in TDA. Mapper is a tool for visualizing high-dimensional data by creating a low-dimensional representation, often in the form of a graph, that preserves the data’s topological structure [Singh et al., 2007]. The Mapper algorithm works by covering the data with overlapping clusters, which are then connected based on their shared data points. This process results in a graph that can reveal the shape and connectivity of data clusters, offering a holistic view of data structure. In financial contexts, Mapper can identify groups of borrowers with similar characteristics or risk levels, providing a basis for segmentation and targeted analysis.

2.3.2 Advantages of TDA

One of the main advantages of TDA is its resilience to noise and ability to capture complex data structures that other methods might overlook. By focusing on topological features rather than point-wise relationships, TDA is less sensitive to noise, making it ideal for financial datasets that often contain irregularities or outliers. Moreover, TDA does not require explicit feature engineering; instead, it uses the data’s geometric properties to generate features, reducing the risk of information loss during preprocessing [Carlsson, 2009]. These attributes make TDA a valuable tool for analyzing non-linear, high-dimensional financial data, where traditional models may fall short.

The application of TDA to data science is still a growing area, but recent research indicates its potential in domains that involve complex datasets with hidden structures. The increasing interest in TDA within fields such as genomics, image analysis, and neuroscience underscores its versatility and effectiveness in capturing intricate data patterns. However, while TDA has gained traction in scientific and theoretical contexts, its application in finance, particularly in loan default prediction, remains limited. This study addresses this gap by integrating TDA with machine learning models to explore its utility in financial prediction tasks.

Chapter 3

Research Methodology

3.1 Research Design

This study adopts a comparative research design to evaluate the impact of integrating Topological Data Analysis (TDA) with traditional machine learning models for loan default prediction. The design combines elements of both experimental and observational research, as it assesses the performance of models with and without TDA-derived features across two financial datasets. By comparing traditional machine learning approaches to TDA-enhanced models, this research aims to determine whether TDA can provide a measurable improvement in predictive accuracy, interpretability, and robustness.

The research process is organized into several key stages. First, baseline machine learning models, including logistic regression, knn, svm, random forest, and gradient-boosted decision trees (GBDT), are developed using traditional financial features. These models serve as the control group, establishing a performance benchmark for loan default prediction without the integration of TDA. The second stage involves applying TDA techniques, specifically persistent homology, to generate topological features from the datasets. These TDA-derived features capture complex, non-linear relationships within the data, which are then used to train TDA-enhanced versions of the machine learning models.

In this design, the traditional and TDA-enhanced models are compared on several performance metrics, including accuracy, precision, recall, and f1-score. This comparative approach not only assesses the predictive accuracy of TDA-enhanced models but also examines their interpretability and computational efficiency relative to traditional models. The study hypothesizes that TDA-enhanced models will outperform traditional models, particularly in cases where the data exhibits complex, non-linear patterns that TDA can uniquely capture. This hypothesis will be tested across two datasets, the Default of Credit Card Clients dataset and the Statlog German Credit dataset, allowing for an analysis of TDA's effectiveness across different data structures and scales.

3.2 Data Collection and Description

This study utilizes two datasets, both widely recognized in loan default prediction research: the **Default of Credit Card Clients** dataset and the **Statlog German Credit** dataset. Each dataset presents unique data structures and attributes that are well-suited for evaluating the effectiveness of TDA-enhanced models in predicting loan defaults.

3.2.1 Default of Credit Card Clients Dataset

This dataset, sourced from the UC Irvine Machine Learning Repository, consists of 30,000 instances of credit card clients from Taiwan, with the aim of predicting whether a client will default on their payments in the following month. The dataset includes 23 attributes that cover both demographic information and financial metrics, such as credit limit, payment history, and bill amounts over a six-month period. The target variable, **default payment next month**, is a binary classification variable where “1” indicates a default and “0” represents non-default. Key variables include:

- **Limit Balance (X1):** Total credit amount granted to the client.
- **Payment History (X6–X11):** A sequence of variables that reflect the repayment status for each of the six months prior to the default prediction month.

- **Bill Statement Amounts (X12–X17):** Monthly bill statement amounts for the previous six months.
- **Previous Payment Amounts (X18–X23):** Historical payment amounts for the preceding six months.

This dataset’s structure, where certain features represent multiple months, presents unique challenges for traditional models, which are often limited in capturing temporal relationships. For this reason, the dataset is particularly suitable for TDA, as persistent homology can analyze the data’s topological structure across multiple variables, potentially uncovering patterns indicative of default risk. A full list of the datasets features and description is seen in Table 1.

Variable Name	Description	Details
X1	LIMIT_BAL	Amount of the given credit (NT dollar). It includes both the individual consumer credit and his/her family (supplementary) credit.
X2	SEX	Gender (1 = male; 2 = female).
X3	EDUCATION	Education (1 = graduate school; 2 = university; 3 = high school; 4 = others).
X4	MARRIAGE	Marital status (1 = married; 2 = single; 3 = others).
X5	AGE	Age (year).
X6	PAY_0	History of past payment. The repayment status in September, 2005.
X7	PAY_2	History of past payment. The repayment status in August, 2005.
X8	PAY_3	History of past payment. The repayment status in July, 2005.
X9	PAY_4	History of past payment. The repayment status in June, 2005.
X10	PAY_5	History of past payment. The repayment status in May, 2005.
X11	PAY_6	History of past payment. The repayment status in April, 2005.
X12	BILL_AMT1	Amount of bill statement (NT dollar) in September, 2005.
X13	BILL_AMT2	Amount of bill statement (NT dollar) in August, 2005.
X14	BILL_AMT3	Amount of bill statement (NT dollar) in July, 2005.
X15	BILL_AMT4	Amount of bill statement (NT dollar) in June, 2005.
X16	BILL_AMT5	Amount of bill statement (NT dollar) in May, 2005.
X17	BILL_AMT6	Amount of bill statement (NT dollar) in April, 2005.
X18	PAY_AMT1	Amount of previous payment (NT dollar) in September 2005.
X19	PAY_AMT2	Amount of previous payment (NT dollar) in August 2005.
X20	PAY_AMT3	Amount of previous payment (NT dollar) in July 2005.
X21	PAY_AMT4	Amount of previous payment (NT dollar) in June 2005.
X22	PAY_AMT5	Amount of previous payment (NT dollar) in May 2005.
X23	PAY_AMT6	Amount of previous payment (NT dollar) in April 2005.
Default Payment Next Month	CLASS	Target variable our model training process tries to predict.

Table 3.1: Default Credit Card Client Dataset Features and Description.

The measurement scale for the repayment status is: -1 = pay duly; 1 = payment delay for one month;

2 = payment delay for two months; . . . ; 8 = payment delay for eight months; 9 = payment delay for nine months and above.

3.2.2 Statlog German Credit Dataset

This dataset is also sourced from the UC Irvine Machine Learning Repository. The Statlog German Credit dataset comes in 2 forms:

1. **Statlog German Credit Default Dataset:** This form of the Statlog German Credit dataset is the original dataset. It contains both categorical and numerical features as they relate to the Statlog German Credit dataset. This dataset is made up of 1,000 instances and 20 attributes related to loan applications in Germany.
2. **Statlog German Credit Numeric Dataset:** The Statlog German Credit Numeric dataset is the encoded version of the original Statlog German Credit dataset and is also provided in the UC Irvine Machine Learning Repository. This dataset is made to include only numerical features after feature encoding of categorical features. It is made up of 1,000 instances and 24 attributes. This dataset is used in this research to represent the Statlog German Credit dataset.

The data is structured as a binary classification problem, with a target variable that categorizes applicants as “good” or “bad” credit risks. This dataset provides various demographic and financial features, including:

- **Status of Checking Account:** Indicates the balance and status of an applicant’s checking account.
- **Credit History:** Reflects the applicant’s historical credit performance, categorized into levels indicating positive or negative credit history.
- **Credit Amount:** The loan amount requested by the applicant.
- **Duration in Months:** Length of the loan term in months.
- **Purpose:** The stated purpose of the loan, such as for car purchase or education.

The Statlog German Credit Numeric dataset, is useful for understanding how TDA performs on datasets smaller in scale. Its relatively balanced target distribution (700 instances labeled “good” credit and 300 labeled “bad” credit) allows for reliable model training and evaluation without extensive class balancing techniques. Additionally, the dataset’s mix of demographic and financial indicators provides a good way for examining TDA’s capacity to capture hidden data structures related to creditworthiness. A full list of the datasets features index name as it relates to the original Statlog German Credit Numeric dataset is seen in Table 2.

Index	Variable Name
Index 0	Status of Existing Checking Account
Index 1	Duration
Index 2	Credit History
Index 3	Credit Amount
Index 4	Savings Account/Bonds
Index 5	Present Employment Since
Index 6	Personal Status and Sex
Index 7	Present Residence Since
Index 8	Property
Index 9	Age
Index 10	Other Installment Plans
Index 11	Number of Existing Credits at this Bank
Index 12	Number of People Being Liable to Provide Maintenance For
Index 13	Telephone
Index 14	Foreign Worker
Index 15	Unknow Dummy Variable
Index 16	Unknow Dummy Variable
Index 17	Other Debtors/Guarantors (Dummy Variable 1)
Index 18	Other Debtors/Guarantors (Dummy Variable 2)
Index 19	Housing (Dummy Variable 1)
Index 20	Housing (Dummy Variable 2)
Index 21	Job (Dummy Variable 1)
Index 22	Job (Dummy Variable 2)
Index 23	Job (Dummy Variable 3)
Index 24	Class

Table 3.2: Statlog German Credit Numeric Dataset Features and Description.

These datasets provide a robust foundation for evaluating whether TDA-enhanced models can improve the predictive performance of traditional machine learning techniques in loan default prediction.

3.3 Data Preprocessing

Data preprocessing is a critical step in any machine learning workflow. It involves preparing the dataset by handling missing values, scaling features, encoding categorical variables, and other transformations. These steps ensure the data is suitable for modeling and contribute to the success of the machine learning pipeline.

3.3.1 Removing Irrelevant Features

In the Default of Credit Card Clients dataset, the feature *ID* was removed as it serves only as an identifier. Since it does not contain meaningful information relevant to predicting the target variable, it was excluded from the dataset.

For the Statlog German Credit Numeric dataset, no features were deemed irrelevant or removed as all features provide potentially useful information.

3.3.2 Handling Missing Values

Both the Default of Credit Card Clients dataset and the Statlog German Credit Numeric dataset consist of 24 columns and contain no missing values. A thorough check confirmed the completeness of both datasets, and no further handling of missing values was necessary.

3.3.3 Feature Engineering

Feature engineering involves creating new features or transforming existing ones to improve model performance and better capture the underlying patterns in the data.

In the Default of Credit Card Clients dataset, two new features were added by applying a logarithmic transformation to the *Age* and *Balance Limit* features. Logarithmic transformations are particularly beneficial for features with skewed distributions, as they reduce the effect of extreme values and normalize the distribution. This helps machine learning algorithms, which often perform better on data with less skewed or more Gaussian-like distributions, to learn patterns more effectively. For instance:

- *Age*: Log-transforming *Age* can emphasize smaller changes in younger ages while reducing the influence of larger age values, capturing nonlinear relationships more effectively.
- *Balance Limit*: Applying a logarithmic transformation to *Balance Limit* can smooth out large variations in credit limits, enabling the model to generalize better.

In the Statlog German Credit Numeric dataset, three new features were created by log-transforming the *Age*, *Duration*, and *Credit Amount* features. Similarly:

- *Age*: As in the previous dataset, this transformation mitigates the impact of outliers and captures more subtle relationships.
- *Duration*: Log-transforming *Duration* helps reduce the impact of very long or short durations, especially if the distribution is heavily right-skewed, improving the model's ability to learn meaningful patterns.
- *Credit Amount*: This feature often has a broad range of values; a logarithmic transformation can make the scale more uniform, enhancing the model's robustness and reducing the impact of large credit amounts.

In both datasets, these transformations help to address skewness, stabilize variance, and reveal hidden patterns, ultimately improving model performance.

3.3.4 Encoding Categorical Features

Both datasets used in this analysis consist entirely of numerical features. For the Statlog German Credit Numeric dataset, some of these numerical features represent dummy variables that have been pre-encoded from original categorical features. Since the dataset has already undergone encoding and contains no remaining categorical features, no additional encoding was required.

Similarly, the Default of Credit Card Clients dataset comprises entirely numerical features, eliminating the need for any further categorical encoding.

3.3.5 Feature Scaling

Feature scaling is a critical step in data preprocessing, especially when working with machine learning algorithms sensitive to the scale of input features. Scaling ensures that all features contribute equally to the model training process, preventing features with larger ranges from disproportionately influencing the model's predictions. This is particularly important for distance-based algorithms like k-nearest neighbors (KNN) or gradient-based methods such as logistic regression and support vector machines.

In this study, the *MinMaxScaler* technique was applied to normalize all features in the dataset. *MinMaxScaler* rescales each feature to a range of 0 to 1 by subtracting the minimum value of the feature and dividing it by the feature's range. This transformation ensures that all features are on the same scale while preserving the relationships and distribution of the original data.

Scaling of the features in the Default of Credit Cards Client and Statlog German Credit Numeric dataset helped to improve:

- **Model Training Stability:** This enables the models explored in this study to converge more efficiently, enhancing both their training speed and overall reliability.
- **Prediction Consistency:** Features were treated equally during prediction, avoiding bias toward larger magnitude features.
- **Comparability Across Models:** Ensuring that scaled data was used consistently allowed fair comparisons across different models, each benefiting equally from the scaled input.

MinMax Scaling:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Where:

- X : The original value.
- $X_{\min}$: The minimum value in the feature.
- $X_{\max}$: The maximum value in the feature.
- X_{scaled} : The scaled value, which typically lies in the range $[0, 1]$.

3.3.6 Feature Selection

Feature selection is a critical step in the preprocessing pipeline, aimed at identifying the most relevant features that contribute significantly to the model's predictive performance. In this study, a statistical approach for feature selection using the *SelectFpr* method from the *sklearn.feature_selection* module, which is based on the F-test (*ANOVA*) for classification tasks was used on both datasets.

The *SelectFpr* method evaluates each feature individually by comparing it to the target variable and calculating the feature's statistical significance. This approach is particularly effective in reducing dimensionality, eliminating redundant or less informative features, and potentially improving the model's performance by focusing on the most relevant predictors.

Index	Features	Score	P-Value
5	PAY_0	2877.87	0.0
6	PAY_2	1873.99	0.0
7	PAY_3	1476.69	0.0
8	PAY_4	1267.64	0.0
9	PAY_5	1084.3	0.0
10	PAY_6	912.58	0.0
24	Log_LIMIT_BAL	798.37	0.0
0	LIMIT_BAL	629.68	0.0
17	PAY_AMT1	122.88	0.0
18	PAY_AMT2	119.16	0.0
21	PAY_AMT5	89.99	0.0
20	PAY_AMT4	82.25	0.0
19	PAY_AMT3	79.09	0.0
22	PAY_AMT6	67.36	0.0
1	SEX	43.57	0.0
2	EDUCATION	16.06	0.0
3	MARRIAGE	14.1	0.0
11	BILL_AMT1	8.27	0.0
12	BILL_AMT2	4.84	0.03
13	BILL_AMT3	4.19	0.04
14	BILL_AMT4	2.66	0.1
4	AGE	2.06	0.15
15	BILL_AMT5	1.4	0.24
16	BILL_AMT6	1.22	0.27
23	Log_AGE	0.2	0.65

Table 3.3: Default Credit Card Client Dataset Feature Selection Table.

With an alpha of 0.05, the feature selection process on the default of credit card client data removes the features; BILL_AMT4, AGE, BILL_AMT5, BILL_AMT6, and Log_AGE as they are considered *not statistically significant*.

Index	Features	Score	P-Value
0	Status of existing checking account	109.48	0.0
2	Credit history	48.34	0.0
25	Log_Duration	39.27	0.0
1	Duration	37.7	0.0
4	Savings account/bonds	22.89	0.0
3	Credit amount	18.43	0.0
5	Present employment since	12.79	0.0
10	Other installment plans	12.68	0.0
20	Housing_A152	12.11	0.0
8	Property	11.85	0.0
26	Log_Age	11.78	0.0
9	Age	10.35	0.0
16	Unknown_dummy2	10.0	0.0
15	Unknown_dummy1	9.99	0.0
19	Housing_A151	8.97	0.0
24	Log_Credit amount	6.83	0.01
6	Personal status and sex	6.25	0.01
18	Other_debtors_or_guarantors_A102	5.97	0.01
14	Foreign worker	3.77	0.05
13	Telephone	2.04	0.15
11	Number of existing credits at this bank	1.15	0.28
17	Other_debtors_or_guarantors_A101	0.82	0.37
22	Job_A172	0.73	0.39
7	Present residence since	0.23	0.63
12	Number of people being liable to provide maintenance for	0.11	0.74
21	Job_A171	0.1	0.76
23	Job_A173	0.0	0.97

Table 3.4: Statlog German Credit Numeric Dataset Feature Selection Table.

The features; Foreign worker, Telephone, Number of existing credits at this bank, Other debtors or guarantors A101, Job A172, Present residence since, Number of people being liable to provide maintenance for, Job A171, Job A173 are removed from the Statlog German Credit Numeric dataset and considered *statistically insignificant* in predicting the target variable.

3.3.7 Handling Class Imbalance

Class imbalance is a common issue in loan default prediction, as the proportion of defaulters (minority class) is often significantly smaller than that of non-defaulters (majority class). Imbalanced datasets can lead to biased machine learning models that favor the majority class, resulting in poor predictive performance for the minority class. To address this issue, this study employs the Adaptive Synthetic Sampling (ADASYN) technique, a synthetic oversampling method designed to balance class distributions by generating synthetic samples for the minority class.

ADASYN Algorithm: ADASYN is an advanced oversampling technique that generates synthetic data points for the minority class based on the density of minority class samples in the feature space [He et al., 2008]. Unlike traditional oversampling methods, such as SMOTE, ADASYN adapts the number of synthetic samples generated for each minority class instance based on its difficulty to classify. This approach not only balances the class distribution but also ensures that the model is better equipped to distinguish between classes in areas of feature space with higher complexity.

Class Balancing for Statlog German Credit Dataset: Initial exploratory data analysis (EDA) of the training dataset revealed a significant imbalance in class distribution, with 240 instances of people who defaulted on loan payments compared to 560 instances of those who did not. After applying ADASYN, the dataset was balanced, resulting in 544 instances of defaulters and 560 instances of non-defaulters. This balanced distribution ensures that the models are trained on a dataset that adequately represents both classes, improving their ability to predict defaults accurately.

Class Balancing for Default of Credit Card Clients Dataset: The Default of Credit Card Clients dataset exhibited an even more pronounced imbalance in the training data, with 5,304 instances of defaulters and 18,668 instances of non-defaulters. Following the application of ADASYN, the minority class was augmented, resulting in 19,253 instances of defaulters and 18,668 instances of non-defaulters. This balancing process minimizes the risk of model bias toward the majority class and enhances the model’s capacity to identify defaulters effectively.

3.4 Model Selection and Algorithms

The selection of machine learning models is critical to evaluating the predictive power of both traditional and TDA-enhanced approaches. This study employs five models: Support Vector Machines (SVM), k-Nearest Neighbors (KNN), XGBoost (XGB), Logistic Regression, and Random Forest. These models were chosen based on their established effectiveness in binary classification tasks, their complementary strengths, and their ability to handle the diverse attributes and complexities of the datasets used in this research.

3.4.1 Support Vector Machines (SVM)

Support Vector Machines are widely used for binary classification tasks due to their ability to find an optimal hyperplane that separates classes in high-dimensional feature space [Cortes and Vapnik, 1995]. SVM is particularly effective for datasets with clear class boundaries, as it maximizes the margin between data points of different classes. For loan default prediction, SVM can capture complex relationships between financial and demographic features, especially when kernel functions are employed to model non-linear patterns. However, SVM may struggle with large datasets like the Default of Credit Card Clients dataset due to its computational intensity.

3.4.2 k-Nearest Neighbors (KNN)

The k-Nearest Neighbors algorithm is a non-parametric model that classifies instances based on the majority class of their nearest neighbors [Altman, 1992]. KNN is intuitive and performs well when patterns are localized within the feature space. For this research, KNN is applied as a baseline model to explore its sensitivity to TDA-enhanced features, as these features may help in distinguishing complex data structures. Despite its simplicity, KNN can face challenges with high-dimensional datasets due to the “curse of dimensionality,” where distances between points become less meaningful.

3.4.3 XGBoost (XGB)

XGBoost, an advanced implementation of Gradient Boosted Decision Trees (GBDT), is renowned for its high accuracy and efficiency in structured data tasks [Chen and Guestrin, 2016]. XGB iteratively builds decision trees, optimizing for errors made by previous trees, which allows it to effectively model complex interactions between variables. Its regularization techniques prevent overfitting, making it a robust choice for loan default prediction. XGB’s scalability and ability to handle both categorical and numerical data make it ideal for the Default of Credit Card Clients dataset, while its interpretability tools, such as feature importance scores, provide insights into key predictors of loan default.

3.4.4 Logistic Regression

Logistic regression is a classic statistical model widely used in loan default prediction due to its simplicity, interpretability, and effectiveness in binary classification tasks. By modeling the relationship between predictor variables and the log-odds of the target variable, logistic regression provides easily interpretable coefficients that highlight the relative importance of each feature [Brown and Mues, 2012]. While logistic regression may not perform as well as more complex models on non-linear relationships, its interpretability and computational efficiency make it a valuable baseline for this research.

3.4.5 Random Forest

Random Forest is an ensemble learning model that builds multiple decision trees and aggregates their predictions to improve accuracy and robustness [Breiman, 2001]. It is particularly effective in handling

high-dimensional datasets and capturing non-linear relationships, which are common in financial data. Random Forest’s built-in feature importance metrics provide valuable insights into the contribution of individual features, making it a strong candidate for evaluating TDA-enhanced models. Additionally, its flexibility with missing data and outliers makes it well-suited for real-world financial datasets like the Statlog German Credit dataset.

3.4.6 Justification for Model Selection

The inclusion of these five models allows for a comprehensive evaluation of traditional and TDA-enhanced approaches. Each model brings unique strengths to the analysis:

- SVM and XGB excel in capturing complex patterns, making them ideal for testing TDA’s ability to enhance predictive power.
- KNN provides a straightforward method to assess how TDA features influence local patterns in the data.
- Logistic Regression offers a benchmark for interpretability, ensuring that the inclusion of TDA-derived features remains accessible to financial practitioners.
- Random Forest balances accuracy, interpretability, and robustness, making it a versatile model for comparative analysis.

By using these models consistently across both datasets, this study aims to assess the generalizability of TDA-enhanced features and their impact on predictive performance. The comparative analysis of these models will provide insights into whether TDA offers significant improvements over traditional feature engineering techniques.

3.5 Hyperparameter Tuning

Hyperparameter tuning is a critical step in machine learning, as it involves optimizing the parameters of a model to try to achieve the best possible performance on a given dataset. Unlike model parameters, which are learned during training, hyperparameters are pre-defined settings that influence how a model learns and generalizes. This study uses grid search with cross-validation to identify the optimal hyperparameter configurations for each model, ensuring a fair and robust comparison between the traditional and TDA-enhanced models.

The hyperparameter tuning process was conducted separately for the two datasets—Statlog German Credit Numeric dataset and Default of Credit Card Clients dataset—due to differences in data size and complexity. Below, the hyperparameter grids for each dataset are described in detail.

3.5.1 Statlog German Credit Numeric Dataset

For the Statlog German Credit Numeric dataset, the hyperparameter grids included a wide range of options to explore various configurations. These grids were designed to account for the smaller dataset size and the potential for overfitting, particularly in complex models like SVM and XGBoost. The hyperparameter grids used can be seen in Table 5.

Machine Learning Algorithm	Hyperparameters	Range of Values
SVM	C kernel degree gamma	0.1, 1, 10, 100 linear, rbf, poly, sigmoid 2, 3, 4 scale, auto, 0.001, 0.01, 0.1, 1
KNN	n_neighbors weights p leaf_size algorithm	3, 5, 7, 10 uniform, distance 1, 2 10, 20, 30, 50 auto, ball_tree, kd_tree, brute
XGBoost	n_estimators learning_rate max_depth	50, 100, 200 0.01, 0.1, 0.2 3, 5, 7
Logistic Regression	C solver penalty max_iter	0.01, 0.1, 1, 10, 100 liblinear, lbfgs, sag, saga, newton-cg l1, l2, elasticnet, none 100, 200, 500
Random Forest	n_estimators max_depth min_samples_split	50, 100, 200 3, 5, 10, None 2, 5, 10

Table 3.5: Statlog German Credit Numeric Dataset Hyperparameter Grid Table.

3.5.2 Default of Credit Card Clients Dataset

For the Default of Credit Card Clients dataset, the hyperparameter grids were adjusted to accommodate the larger dataset size, focusing on configurations that are computationally feasible while still providing sufficient variability for optimization. The hyperparameter grids used can be seen in Table 6.

Machine Learning Algorithm	Hyperparameters	Range of Values
SVM	C kernel gamma	0.1, 1, 10 linear, rbf scale, auto
KNN	n_neighbors weights p	3, 5, 7 uniform, distance 1, 2
XGBoost	n_estimators learning_rate max_depth	50, 100, 200 0.01, 0.1, 0.2 3, 5, 7
Logistic Regression	C solver	0.01, 1, 10 liblinear, lbfgs
Random Forest	n_estimators max_depth min_samples_split	50, 100, 200 3, 5, 10, None 2, 5, 10

Table 3.6: Default of Credit Card Client Dataset Hyperparameter Grid Table.

3.5.3 Grid Search and Cross-Validation

For both datasets, grid search was employed to systematically evaluate all possible combinations of hyperparameters within the specified grids. This process was combined with 5-fold StratifiedKfold cross-validation to ensure that the models were trained and validated on balanced subsets of the data. StratifiedKfold preserves the proportion of classes (e.g., defaulters and non-defaulters) in each fold, which is especially important for imbalanced datasets like those used in this study. By maintaining consistent class distributions across folds, StratifiedKfold reduces the risk of biased performance estimates, leading to more reliable evaluation results.

3.5.4 Outcome of Hyperparameter Tuning

The optimal hyperparameter configurations for each model were determined based on their performance on cross-validation metrics, specifically the F1-score. The F1-score balances precision and recall to provide a single metric that accounts for both false positives and false negatives. The F1-score is particularly relevant in loan default prediction, where minimizing false negatives (misclassifying defaulters as non-defaulters) is critical for effective risk management. These configurations were then used to train the final models, which were subsequently evaluated on the test data to assess their predictive performance.

3.5.5 StratifiedKFold Cross-Validation

StratifiedKFold was chosen as the cross-validation strategy for its ability to maintain class balance in each fold. This approach divides the dataset into k folds, ensuring that each fold preserves the class distribution of the original dataset. For this study, 5-fold StratifiedKFold was applied to both datasets during grid search, ensuring that the minority class (defaulters) was proportionately represented in all training and validation splits. The process followed these steps:

1. **Dataset Splitting:** The dataset was divided into five equal-sized folds, each containing the same proportion of defaulters and non-defaulters as the overall dataset.
2. **Model Training and Validation:** For each combination of hyperparameters, the model was trained on four folds and validated on the remaining fold, cycling through all five folds to complete the cross-validation process.
3. **Performance Aggregation:** The F1-scores from all five folds were averaged to evaluate the overall performance of the model for the given hyperparameter configuration.

3.5.6 Hyperparameter Tuning Steps

The following steps were taken across both datasets for hyperparameter tuning:

1. Get the dataset.
2. Extract the data features and data label.
3. Split the data features and data label into a training set(80%) and test set(20%).
4. Create a dictionary of the models and the parameter grids to be explored with grid search.
5. Define a classifier to be passed with the parameter grids into the grid search class.
6. Fit the classifier to the training set.
7. Generate predictions for the test set.
8. Retrieve optimal parameters using `best_estimator_.get_params()`.

3.6 Evaluation Metrics

The performance of predictive models in this study is evaluated using four key metrics: Accuracy, Precision, Recall, and F1-Score. These metrics provide a comprehensive view of model performance, addressing the trade-offs between correctly identifying loan defaulters (minority class) and avoiding false alarms. The following notations below will be used in providing context for each evaluation metric

- TP (True Positives): Correctly predicted defaulters.
- TN (True Negatives): Correctly predicted non-defaulters.
- FP (False Positives): Non-defaulters are incorrectly predicted as defaulters.
- FN (False Negatives): Defaulters are incorrectly predicted as non-defaulters.

3.6.1 Accuracy

Accuracy measures the proportion of correctly classified instances out of the total number of instances. While useful for balanced datasets, accuracy can be misleading in imbalanced datasets, as it does not differentiate between the performance on the majority and minority classes. For this reason, accuracy is supplemented with other metrics in this study.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

3.6.2 Precision

Precision evaluates the proportion of correctly predicted positive instances (defaulters) among all instances predicted as positive. It measures the model's ability to avoid false alarms, which is crucial in minimizing the cost of incorrectly classifying non-defaulters as defaulters.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

A high precision indicates a low false positive rate, ensuring that most of the predicted defaulters are actual defaulters.

3.6.3 Recall

Recall, also known as sensitivity or true positive rate, measures the proportion of actual defaulters that the model correctly identifies. It focuses on minimizing false negatives, which is critical in applications like loan default prediction where failing to identify defaulters can result in significant financial losses.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

A high recall indicates that the model effectively captures the majority of defaulters in the dataset.

3.6.4 F1-Score

The F1-Score is the harmonic mean of Precision and Recall, providing a single metric that balances the trade-off between these two measures. It is particularly useful for imbalanced datasets, where both false positives and false negatives have significant implications.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-Score ranges from 0 to 1, where a score closer to 1 indicates a better balance between Precision and Recall.

3.7 TDA Techniques

3.7.1 Topology and Data

Topology, in the context of data analysis, refers to the study of the properties of data that are preserved under continuous transformations, such as stretching or bending, but not tearing or gluing. The fundamental concept of Topological Data Analysis (TDA) is that the shape, or topology, of a dataset can reveal critical insights into its structure. Unlike traditional approaches that focus on pointwise relationships, TDA aims to capture the global structure of the data, providing a higher-level understanding of its intrinsic geometry [Carlsson, 2009].

At the core of TDA is the idea of topological features, such as connected components, loops, and higher-dimensional voids, which describe the shape of the data. These features capture how the data points are organized and related. For example, in high-dimensional datasets, TDA can detect clusters of data points forming meaningful geometric shapes, which may be obscured by noise when analyzed through conventional methods [Zomorodian, 2005]. TDA can also uncover structures in data that remain consistent under small changes, such as "holes" in the data or isolated components.

TDA does not assume any prior knowledge of the data's distribution or geometry, which makes it a powerful tool for identifying complex structures, even in noisy or incomplete datasets [Ghrist, 2008]. This capability allows it to be applied to a wide range of problems, from biological data analysis to financial risk prediction.

3.7.2 Simplex and Simplicial Complex

In Topological Data Analysis (TDA), the concept of a simplex is fundamental. A simplex is the simplest possible geometric object in a given dimension. A simplex refers to a basic geometric object in a given space, such as a triangle in 2D space, a tetrahedron in 3D space, and so on. It can be considered as the generalization of a triangle or tetrahedron to any number of dimensions. For example:

- A 0-simplex is a point.
- A 1-simplex is a line segment, defined by two points.
- A 2-simplex is a triangle, defined by three points.
- A 3-simplex is a tetrahedron, defined by four points.

A simplicial complex is a set of simplices that are combined in a way that satisfies two conditions:

1. **Non-empty intersection:** Any two simplices in the complex either do not intersect, or their intersection is a face of both simplices.
2. **Closure:** Every face of a simplex in the complex is also part of the complex. For a dataset, simplices represent the relationships between data points in terms of proximity or connectivity. The process of constructing simplices from data points involves a filtration process, which progressively builds higher-dimensional simplices as more data points are added to the analysis.

For example, given a point cloud, a 1-skeleton of the data consists of all the edges (1-simplices) formed by pairs of points that are sufficiently close to each other. As more simplices are added (such as triangles and tetrahedra), the complex provides a more detailed representation of the topological features present in the data.

Simplices are crucial in TDA because they form the building blocks of the Vietoris-Rips complex, which is used in the persistent homology process to study the changing topological features of a dataset across multiple scales [Zomorodian and Carlsson, 2005].

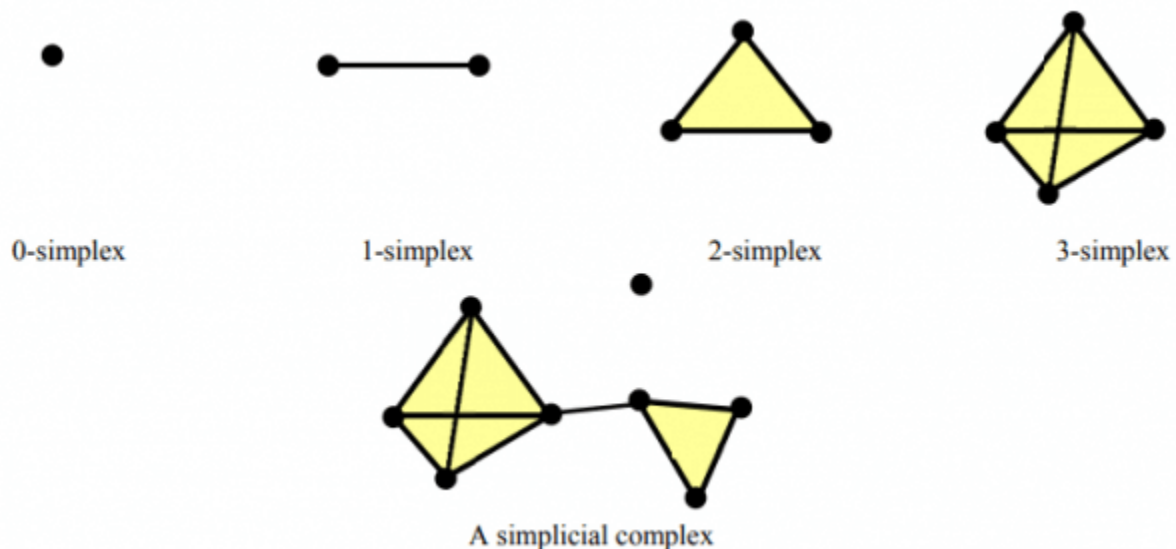


Figure 3.1: Building Simplicial Complexes (Source: [Zulkepli et al., 2019])

3.7.3 Filtration

In Topological Data Analysis (TDA), a filtration is a process of building a sequence of simplicial complexes that captures the topological features of the data at various scales. The filtration provides a way to study how the topological features evolve as we progressively add simplices to the complex. This is crucial for understanding the persistence of features like connected components, loops, and voids in the dataset.

The filtration process can be visualized as a growing network of simplices where, at each step, new simplices are added to reflect the data's increasing connectivity or complexity. This can be done using various methods:

1. Vietoris-Rips Filtration: One of the most common methods, where simplices are added based on the distance between points. In this case, points that are within a certain distance (called a "radius") from each other form edges (1-simplices), and higher-dimensional simplices are formed by progressively larger groups of points. The Vietoris-Rips filtration process is used in the study to create the simplicial complex upon which the barcode statistics and TDA-derived features are built.
2. Cech Filtration: Similar to Vietoris-Rips, but uses balls centered at each point instead of simply connecting points within a given distance threshold.

The filtration process allows for the study of topological features (such as connected components and loops) at various scales. For example, when the radius is small, the data might consist of many small connected components, but as the radius increases, some of these components may merge, revealing larger structures or topological features. This progression is critical for persistent homology, which analyzes the "birth" and "death" of these features over different filtration scales.

The result of the filtration is a set of persistence barcodes or persistence diagrams, which visualize the birth and death of topological features (such as connected components and loops) as the filtration process progresses.

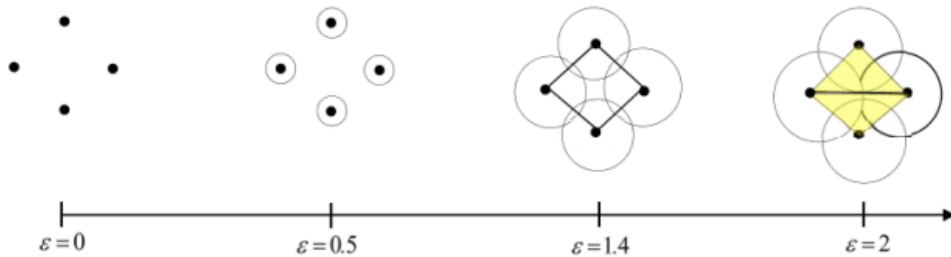


Figure 3.2: Filtration Process at Different Values of Epsilon (Source: [Zulkepli et al., 2019])

3.7.4 Homology Groups

Homology groups are fundamental objects in algebraic topology that capture the topological features of a dataset, such as connected components, loops, and higher-dimensional voids. In the context of Topological Data Analysis (TDA), homology groups formalize the persistence of these features as we construct the simplicial complex during the filtration process. Each homology group H_k corresponds to a different type of topological feature:

- H_0 : Represents **connected components**.
- H_1 : Represents **loops or cycles**.
- H_2 : Represents **voids or 2-dimensional holes**.
- Higher-dimensional homology groups ($H_3, H_4, \dots$) correspond to increasingly complex topological features, such as higher-dimensional cavities or voids.

Interpretation of Homology Groups

- **Connected Components (H_0):** The homology group H_0 is associated with connected components in the data. Intuitively, each connected component represents a separate cluster of points. As the filtration progresses, connected components merge, and the corresponding features in H_0 disappear [Edelsbrunner and Harer, 2010].
- **Loops (H_1):** The homology group H_1 corresponds to loops or cycles in the dataset. A loop is a closed curve that cannot be contracted to a point. For example, a circular or ring-shaped structure in the data corresponds to a feature in H_1 . The *birth* of a loop is recorded when it first appears in the filtration, and its *death* is recorded when it is filled in or merged with other structures [Ghrist, 2008].
- **Voids and Higher-Dimensional Features (H_2 and beyond):** For higher dimensions, homology groups H_2 and beyond capture voids and higher-dimensional features, such as cavities in a 3D point cloud. However, in many practical applications, H_0 and H_1 provide the most relevant information as they correspond to basic features like clusters and loops.

Example in TDA

Consider a dataset represented as a point cloud in $\mathbb{R}^2$. As we increase the radius for the Vietoris-Rips filtration:

- At small scales, small clusters of points form connected components (H_0).
- At intermediate scales, loops or cycles may appear, corresponding to features in H_1 .

These features are tracked across different scales of the filtration, and their persistence is recorded to generate a **persistence diagram** or **barcode**, summarizing the topological features in terms of their birth and death times.

Homology groups provide a rigorous way to quantify the topological features in a dataset. The use of H_0 and H_1 homology groups allows us to focus on the most important features—connected components and loops—that reveal the intrinsic structure of the data. By analyzing the persistence of these features, we can gain insights into the data’s geometry and detect hidden patterns in datasets.

3.7.5 Persistence Homology

Persistence homology is a central concept in Topological Data Analysis (TDA), providing a way to track the birth and death of topological features such as connected components, loops, and voids across different scales. The idea is to capture the evolution of these features as a parameter, typically the radius in a Vietoris-Rips filtration, increases. This technique allows us to identify significant topological features that persist over a wide range of scales, while also filtering out those that appear due to noise.

Persistent homology examines how the homology groups of a simplicial complex change as we build the complex through the filtration process. Specifically, it tracks the birth and death times of topological features in each homology group, which is a key step in understanding the data’s intrinsic structure [Zomorodian and Carlsson, 2005].

Filtration and Persistent Features

As we increase the scale in a filtration, simplices are added, and topological features emerge. For example, as the radius in a Vietoris-Rips filtration increases, small connected components may merge to form larger connected components. These features are tracked by computing the homology of the complex at each step of the filtration.

- Birth time refers to the scale at which a topological feature first appears in the complex.
- Death time refers to the scale at which the feature disappears or merges with another feature.

Persistent homology focuses on the features that persist over a range of scales, as these are considered more significant and less likely to be artifacts of noise. Features that appear and disappear quickly are considered to be less persistent and are typically discarded.

Mathematical Formulation

The formalism of persistent homology is built on the notion of homology groups, which capture topological features like connected components (H_0), loops (H_1), and higher-dimensional voids ($H_2, H_3, \dots$). The persistence of these features is captured in the form of a persistence diagram or a barcode, where each feature is represented as a bar whose length indicates the persistence of the feature (i.e., the difference between its death and birth times).

Let us focus on the two primary homology groups used in persistent homology which are used in this study:

- H_0 : Represents the number of connected components in the dataset. Features in H_0 correspond to clusters of data points [Edelsbrunner and Harer, 2010].
- H_1 : Represents loops in the dataset. Features in H_1 capture circular structures that cannot be contracted to a point [Ghrist, 2008].

The persistence of a feature F in a given homology group H_k is often represented as an interval $[b, d]$, where b is the birth time and d is the death time of the feature. The persistence of the feature is $d - b$, and this value provides a measure of how significant the feature is within the dataset.

3.7.6 Persistence Barcodes

A persistence barcode is a visual representation of the persistence of topological features, such as connected components, loops, and higher-dimensional voids, during the filtration process in Topological Data Analysis (TDA). Barcodes provide an intuitive way to capture the birth and death of these features across various scales, offering valuable insights into the dataset's topological structure.

Representation

In visualizing persistence barcodes:

- Each topological feature is represented as a horizontal bar.
- The **birth** of a feature corresponds to the left endpoint of the bar.
- The **death** corresponds to the right endpoint of the bar.
- The **length of the bar** represents the *persistence* of the feature, defined as the difference between its death and birth times ($d - b$).
- **Longer bars** indicate more persistent, significant features, while **shorter bars** are typically considered noise.

Mathematical Definition

Mathematically, if a feature has birth time b and death time d , the corresponding bar spans from b to d on the x-axis. The persistence of the feature is given by:

$$\text{Persistence} = d - b,$$

quantifying how long the feature remains active during the filtration process.

The persistence barcode serves as a compact summary of the topological features in a dataset, illustrating how these features change as the filtration radius increases.

Interpretation of Persistence Barcodes

- **Long Bars:** Features that persist for a long time across scales are considered important and robust. These bars are associated with topological features that are more likely to represent meaningful structures in the data.
- **Short Bars:** Features that appear and disappear quickly are typically considered noise. These features do not persist across multiple scales and are unlikely to represent significant patterns within the dataset.

The persistence barcode allows us to visually analyze the topological structure of a dataset at different scales, enabling us to distinguish between important and unimportant features. This is especially useful in high-dimensional datasets where traditional methods may fail to capture complex relationships.

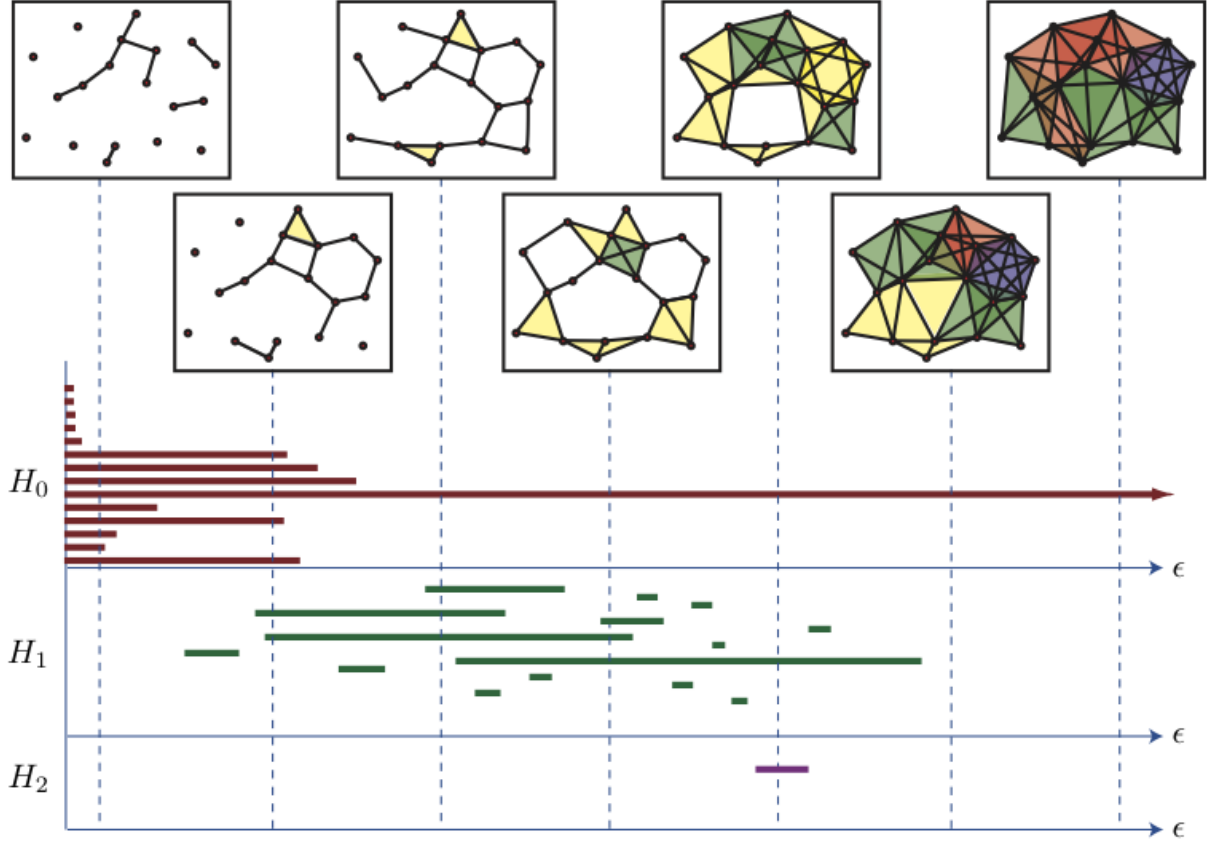


Figure 3.3: Illustration of Filtration Process with Persistence Barcodes Recorded for Connected Components (H_0), Loops (H_1), and Cavities (H_2) at Different Values of Epsilon (Source: [Warrior, 2015])

3.7.7 Barcode Statistics

Barcode statistics are numerical summaries derived from persistence barcodes, providing a quantitative representation of the topological features identified during the filtration process. These statistics are computed for each homology group (H_0, H_1) to capture key attributes such as the birth and death times of features, their persistence, and their distribution across the dataset [Sagming et al., 2024]. By transforming persistence diagrams into fixed-dimensional statistical summaries, barcode statistics enable the integration of topological features into traditional machine learning workflows.

Step 1: Extract Key Metrics

For each persistence diagram, the following metrics are extracted:

- **Birth Time (b):** The scale at which a topological feature (e.g., a connected component or a loop) first appears in the filtration process.
- **Death Time (d):** The scale at which the feature disappears or merges with another feature during the filtration.
- **Persistence ($d - b$):** The difference between the death time and birth time, representing the lifespan of the feature. Persistent features are considered significant, while short-lived features are often attributed to noise.
- **Gap to Maximum Death Time ($y_{\max} - d$):** The difference between the maximum death time ($y_{\max}$) in the diagram and each individual death time. This metric reflects how close a feature is to the largest-scale event in the filtration. (For simplicity, we'll call this the *Residual Death Time*.)

Step 2: Compute Statistical Measures

For each of the four metrics above (birth time, death time, persistence, and residual death time), the following statistical measures are calculated:

- **Mean:** The average value of the metric across all features in the persistence diagram.
- **Median:** The middle value of the metric when sorted, providing a measure less affected by outliers.
- **Standard Deviation:** A measure of the spread or variability of the metric, indicating how much the values deviate from the mean.

Summary of Barcode Statistics

This process results in a total of 12 statistics per persistence diagram:

- For Birth Time (b): Mean, Median, Standard Deviation.
- For Death Time (d): Mean, Median, Standard Deviation.
- For Persistence ($d - b$): Mean, Median, Standard Deviation.
- For Residual Death Time ($y_{\max} - d$): Mean, Median, Standard Deviation.

Metric	Mean	Median	Standard Deviation
Birth Time (b)	g_1	g_5	g_9
Death Time (d)	g_2	g_6	g_{10}
Persistence ($d - b$)	g_3	g_7	g_{11}
Residual Death Time ($y_{\max} - d$)	g_4	g_8	g_{12}

Computed Barcode Statistics

The twelve barcode statistics calculated for each persistence barcode (H_0 and H_1) used in this study are described in Table 3.7.

Statistic	Description
g_1 (Mean Birth)	Average birth time of features.
g_2 (Mean Death)	Average death time of features.
g_3 (Mean Persistence)	Average persistence (death – birth).
g_4 (Mean Gap to Max Death)	Average gap between maximum death time ($y_{\max}$) and individual death times.
g_5 (Median Birth)	Median birth time of features.
g_6 (Median Death)	Median death time of features.
g_7 (Median Persistence)	Median persistence (death – birth).
g_8 (Median Gap to Max Death)	Median gap between $y_{\max}$ and individual death times.
g_9 (Std Birth)	Standard deviation of birth times.
g_{10} (Std Death)	Standard deviation of death times.
g_{11} (Std Persistence)	Standard deviation of persistence (death – birth).
g_{12} (Std Gap to Max Death)	Standard deviation of gaps to $y_{\max}$.

Table 3.7: Computed Barcode Statistics for Persistence Barcodes.

Chapter 4

Implementation

4.1 Development Environment

The implementation of this research was carried out in a structured and well-defined development environment to ensure the smooth execution of tasks related to data preprocessing, topological data analysis (TDA), and machine learning model training and evaluation. Below are the key components of the development environment:

4.1.1 Programming Language

The entirety of the research implementation was performed using Python, a versatile and widely used programming language for data analysis and machine learning tasks. Python was chosen for its extensive library ecosystem, ease of use, and strong community support, making it ideal for handling the diverse requirements of this study.

4.1.2 Libraries and Frameworks

A range of Python libraries was utilized to address various aspects of the research, from data manipulation to advanced machine learning and TDA:

- **Numpy** and **Pandas**: For numerical computations and efficient data manipulation.
- **Matplotlib** and **Seaborn**: For data visualization and exploratory data analysis (EDA).
- **Scikit-learn**: For machine learning model implementation, hyperparameter tuning, and evaluation.
- **XGBoost**: For gradient boosting machine learning models.
- **ydata_profiling**: For generating comprehensive data profiling reports during the EDA phase.
- **Imbalanced-learn (imblearn)**: For handling class imbalance using techniques such as ADASYN and undersampling.
- **Ripser**: For constructing Vietoris-Rips complexes and extracting persistent homology.
- **Persim**: For visualizing persistence diagrams generated during TDA.
- **ucimlrepo**: For accessing datasets, particularly the Statlog German Credit dataset.
- **Joblib**: For saving and loading trained models and intermediate results.
- **Warnings**, **glob**, and **os**: For managing system-level tasks and suppressing unnecessary warnings.

4.1.3 Hardware Specifications

The implementation was conducted on a personal computer with the following specifications:

- **Processor:** Intel(R) Core(TM) i3-10110U CPU @ 2.10GHz 2.59GHz
- **RAM:** 8.00 GB (7.81 GB usable)
- **System Type:** 64-bit operating system, x64-based processor

While the hardware was sufficient for handling most tasks, the creation of landmark sets and computation of Vietoris-Rips complexes proved to be computationally intensive, occasionally pushing the limits of the available RAM and processor capacity. These processes would benefit from higher-end hardware, such as processors with more cores or additional RAM, particularly for large datasets like the Default of Credit Card Clients dataset.

4.1.4 Integrated Development Environment (IDE)

The implementation was carried out using **Spyder**, the latest version bundled with **Anaconda**. Spyder was selected for its user-friendly interface, robust debugging tools, and seamless integration with the scientific Python stack, providing an efficient environment for data analysis and model development.

4.1.5 Version Control

Version control was managed using **GitHub** and **GitHub Desktop**. These tools were employed to track changes in the codebase, ensure proper versioning of scripts and datasets, and maintain a backup of all project files. The use of version control facilitated collaboration, code organization, and reproducibility.

4.1.6 Challenges and Special Considerations

A notable challenge encountered during the implementation was the computational intensity of creating landmark sets and calculating topological features using Ripser. The process, particularly for large datasets like the Default of Credit Card Clients, required substantial computational resources, pushing the limits of the available hardware. This limitation was mitigated by carefully optimizing the code and leveraging the landmark sampling approach to reduce the dataset size without compromising topological fidelity.

4.2 Data Preprocessing Pipeline

The Statlog German Credit dataset and the Default of Credit Card Clients dataset both undergo a systematic cleaning and preprocessing pipeline to prepare them for model building and training. Figure 4.1 provides a high-level overview of the steps involved, while Figure 4.2 offers a more detailed breakdown of the specific preprocessing tasks, along with a visual representation of the data flow through the pipeline for the data mining process without TDA-derived features. The data preprocessing process for each stage is described in Section 3.3.

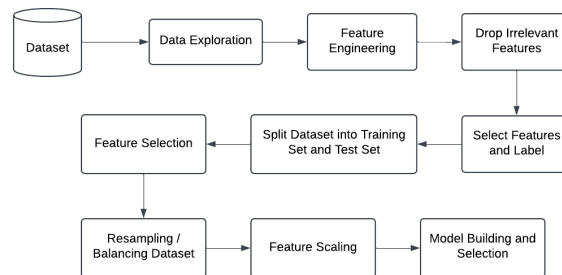


Figure 4.1: Preprocessing Pipeline for the datasets.

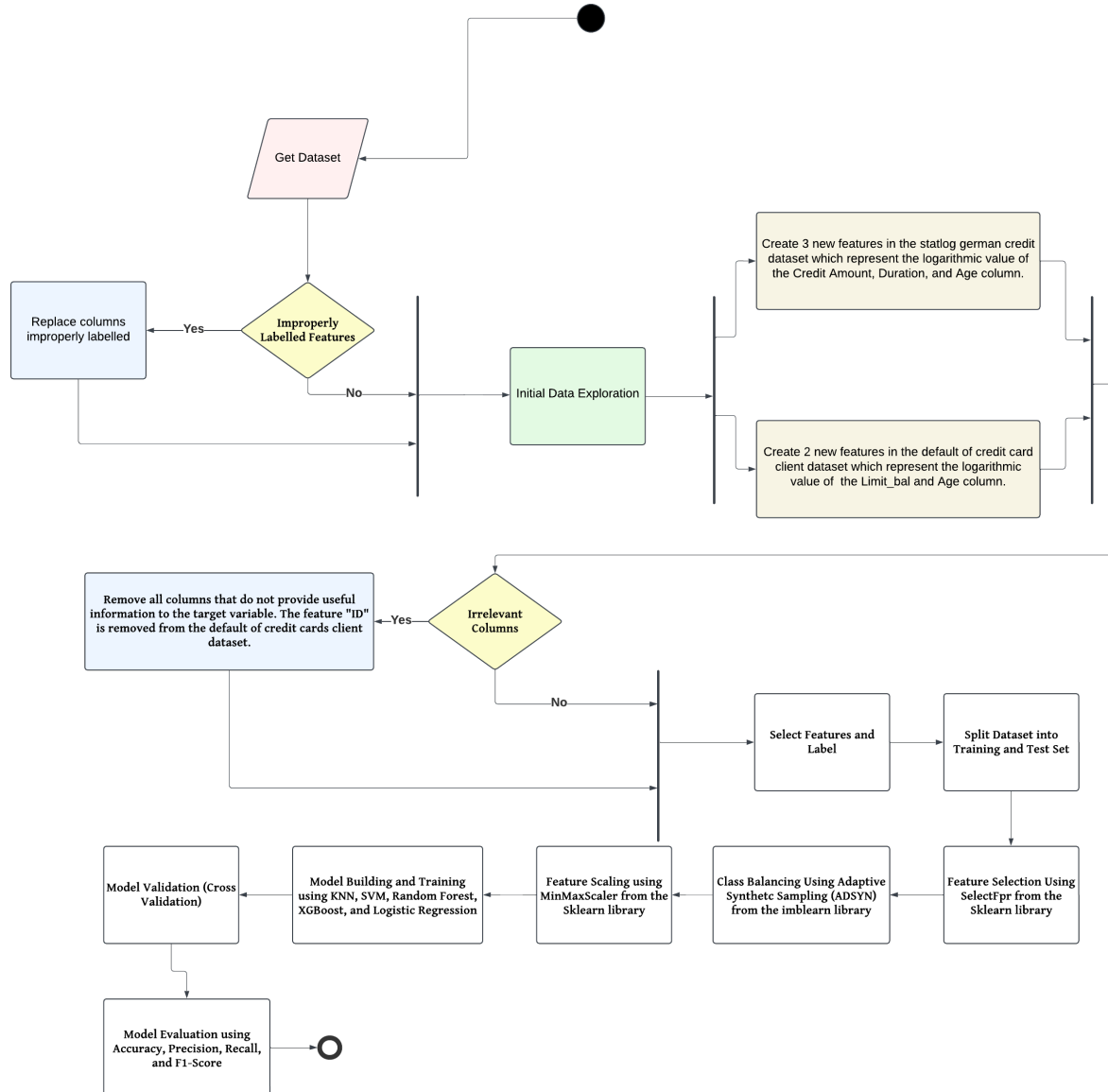


Figure 4.2: Pipeline For Data Mining Without TDA-Derived Features

4.3 Default Prediction System Design

The process for default prediction is illustrated in Figure 4.3 below.

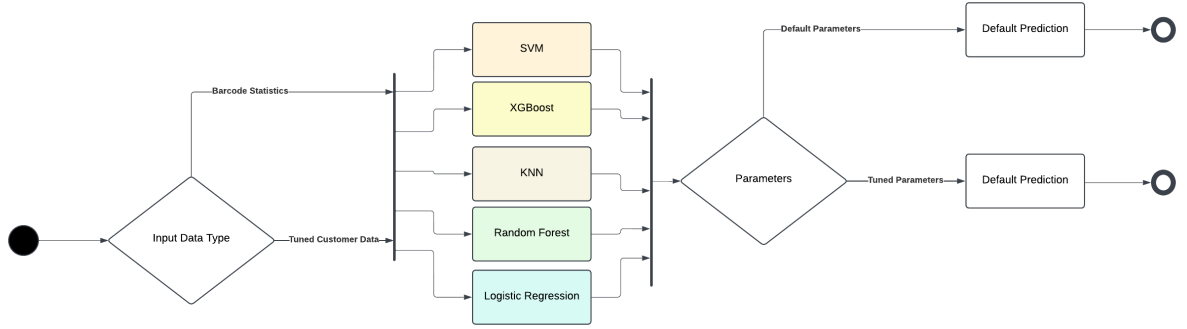


Figure 4.3: Default Prediction Design for the Statlog German Credit dataset and the Default of Credit Cards Client dataset

4.4 Hyperparameter Tuning System Design

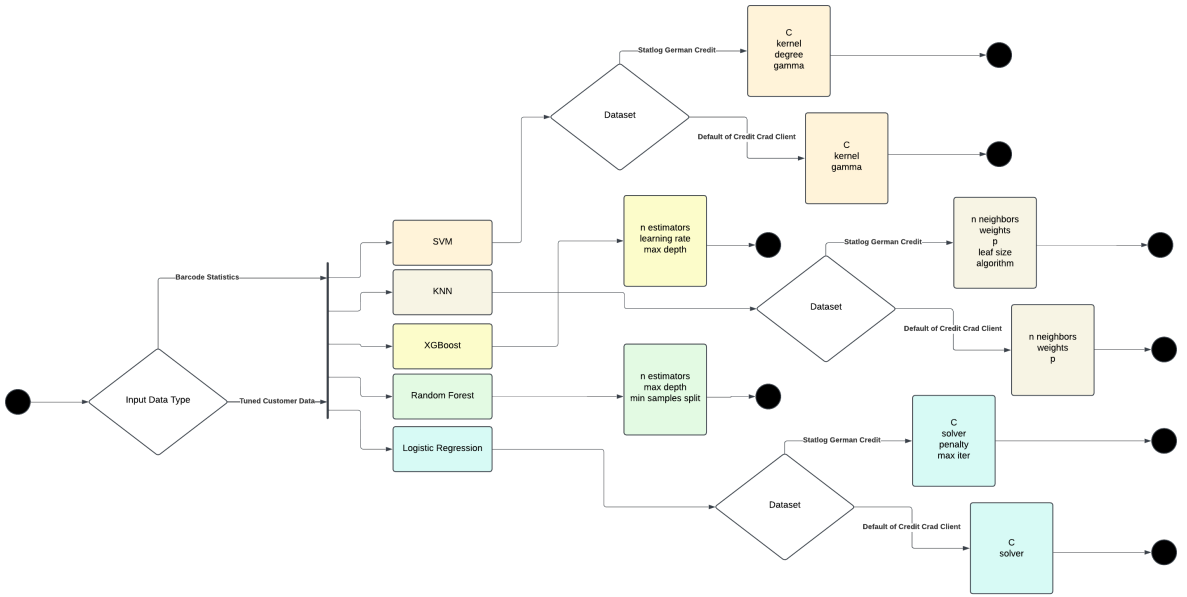


Figure 4.4: Hyperparameter Tuning Process for the Statlog German Credit dataset and the Default of Credit Cards Client dataset

4.5 Selecting Landmarks and Computing Barcode Statistics

With the goal of this research thesis being to apply TDA to machine learning models, we consider two major blocks that help in building this process. These are Landmarks and Barcode Statistics. These two serve as the foundation upon which the topological structure of a dataset can be extracted.

4.5.1 Landmarks

In the context of Topological Data Analysis (TDA), a landmark set refers to a subset of points from the original point cloud data. Point cloud data is a geometric representation of a dataset in a finite metric space, where each data point corresponds to a position in that space [Sagming et al., 2024]. The

purpose of a landmark set is to preserve the essential topological structure of the dataset while reducing computational complexity and enabling efficient analysis of large datasets.

Constructing a landmark set involves selecting representative points that capture the geometry and topology of the original dataset. This is particularly critical in datasets with high dimensionality or large sample sizes, such as the ones used in this study, where analyzing the full dataset would impose significant computational demands. By leveraging landmarks, we can create a simplified yet topological representation of the data, facilitating the extraction of meaningful topological features.

Selection Methods for Landmarks

Several methods exist for selecting landmarks, depending on the dataset’s characteristics and the goals of the analysis. Common approaches include:

- **Max-Min Selection:** This method iteratively selects points that maximize the minimum distance to previously selected landmarks, ensuring diverse coverage of the dataset’s structure [de Silva and Carlsson, 2004].
- **Random Sampling:** A simple yet effective approach where landmarks are chosen randomly, often with stratification to ensure proportional representation of different regions in the dataset. This technique is utilized for our landmark selection in this study.
- **Density-Based Sampling:** Points are selected based on density estimations, favoring areas with greater data concentration, which often represent critical topological features.

Selecting Landmarks for the Dataset

Both the Statlog German Credit dataset and the Default of Credit Card Clients dataset exhibit significant class imbalance, with the majority class (non-defaulters) far outnumbering the minority class (defaulters). To mitigate this issue and ensure equitable representation of both classes, undersampling was applied to both datasets as a preprocessing step prior to selecting landmarks for TDA. Before undersampling, the Statlog German Credit dataset consisted of 300 instances of defaulters and 700 instances of non-defaulters, resulting in a skewed class distribution. To address this, the dataset was undersampled to retain 300 instances from each class, achieving a balanced distribution of 300 defaulters and 300 non-defaulters. For the Default of Credit Card Clients dataset, there was a significant class imbalance, with 6,630 instances of defaulters compared to 23,335 instances of non-defaulters. Following the undersampling process, 6,630 instances were retained from each class, resulting in a balanced dataset of 6,630 defaulters and 6,630 non-defaulters.

After undersampling the majority class for both datasets, 4 landmarks were defined on both datasets for the defaulters and non-defaulters. For the Statlog German Credit dataset, the landmark sizes selected are L_{30} , L_{40} , L_{50} , and L_{60} which represent 30%, 40%, 50%, and 60% of the dataset respectively while for the Default of Credit Card Client dataset, the landmark sizes selected are L_5 , L_{10} , L_{15} , and L_{20} which represent 5%, 10%, 15%, and 20% of the dataset. For each dataset, 500 landmark sets were generated separately for defaulters and non-defaulters for each specified landmark size.

For instance, consider the Statlog German Credit dataset. After undersampling, we begin with a landmark size L_{30} . For customers who defaulted, we randomly selected 30% of the total defaulters based on the specified landmark size. This process is repeated 500 times, generating 500 distinct landmark sets for defaulters. The same procedure is applied to customers who did not default generating a total of 1000 landmark files for L_{30} . The process is then repeated for additional landmark sizes: L_{40} , L_{50} , and L_{60} .

A step-by-step process for the landmark selection is seen below:

1. Import Pandas, Numpy, the preprocessing module from the Sklearn library in Python.
2. Using Pandas, we import the preprocessed dataset.
3. Using the preprocessing module from the Sklearn library, we normalize the preprocessed dataset using the MinMaxScaler.
4. We split the normalized dataset into 2 groups: Defaulters and Non-Defaulters.
5. We perform undersampling on the majority class to balance both groups.

6. Define the landmark sizes for both datasets. For the Statlog German Credit dataset, we specify the landmark sizes L_{30} , L_{40} , L_{50} , and L_{60} , while for the Default of Credit Card Clients dataset, we use the landmark sizes L_5 , L_{10} , L_{15} , and L_{20} .
7. For L_5 , we randomly select 500 landmark sets for defaulters and non-defaulters respectively taking 5% of the dataset. We store each of the landmark sets in different files making a total of 1000 landmark files. We repeat the process for L_{10} , L_{15} , and L_{20} .
8. We repeat Step 7 for the Default of Credit Card Client dataset and generate 1000 landmark files for L_{30} , L_{40} , L_{50} , and L_{60} .

Justification for Choice of Landmark Sizes

For each landmark set, a Vietoris-Rips complex is constructed, and barcode statistics are extracted using the Ripser algorithm. However, applying the Ripser algorithm directly to the entire dataset becomes computationally expensive as dataset size increases. To address this, landmarks are generated to represent the dataset at different sample sizes.

The Statlog German Credit dataset, being relatively small, has 300 instances of defaulters and 300 instances of non-defaulters after undersampling. With a landmark size of 5%, each landmark set contains 15 samples per class. This small sample size may not accurately capture the topological structure of the classes, especially considering the high dimensionality of the dataset, where the number of features exceeds the number of samples. To address this limitation, larger landmark sizes are used to better represent the topological structure of the dataset.

In contrast, the Default of Credit Card Clients dataset, with 6,630 instances of defaulters and non-defaulters after undersampling, benefits from a 5% landmark size. This produces 331 samples per landmark set for each class, which is sufficient to capture the topological structure of the data. The larger dataset size allows for meaningful topological insights even with smaller landmark percentages, making 5% an appropriate choice for this dataset.

4.5.2 Barcode Statistics

After generating the landmark sets for the datasets, barcode statistics are extracted by constructing a simplicial complex using the Vietoris-Rips filtration process. This process enables the extraction of key topological features from the data, as defined by homology groups [Sagming et al., 2024]. In this study, the H_0 and H_1 homology groups were utilized, representing the number of connected components (0-dimensional holes) and loops (1-dimensional holes), respectively, as observed during the filtration process.

For each landmark size in both datasets, a persistence barcode was computed to capture the birth and death of topological features during the filtration process, separately for defaulters and non-defaulters. These diagrams provide insights into the topological structures associated with each class, offering a deeper understanding of the dataset’s geometric and structural properties.

The steps involved in computing the barcode statistics for each landmark set are as follows:

1. Import the required libraries: Pandas, NumPy, and Ripser in Python.
2. For each landmark file L_{30} , L_{40} , L_{50} , and L_{60} or L_5 , L_{10} , L_{15} , and L_{20} , extract the individual landmark sets.
3. Construct a Vietoris-Rips complex for each landmark set to capture its topological information. This is performed using the Ripser library.
4. Extract the H_0 and H_1 homology groups from the simplicial complex.
5. From H_0 and H_1 , retrieve the birth times and death times of each topological feature. Replace infinity values with 22, as barcode points are normalized such that the maximum distance between two points is the diagonal of a unit square.
6. Compute 12 barcode statistics for H_0 and another 12 for H_1 , and store the results in an output matrix. Append the associated label for the landmark set at the end of the matrix, where defaulters are labeled as 1 and non-defaulters as 0.
7. Repeat steps (2) to (6) for all defaulters and non-defaulters within the current landmark file.

8. Iterate steps (2) to (7) for all landmark files.
9. Combine the results into a final array of size 1000×25 , where the rows represent the barcode statistics for 500 landmark sets from defaulters and 500 from non-defaulters for the specified landmark file. The columns include 12 barcode statistics from H_0 , 12 from H_1 , and the associated class label, totaling 25 columns.
10. Randomize the final results to create the TDA-derived features using the barcode statistics.

4.6 TDA and Barcode Statistics System Design

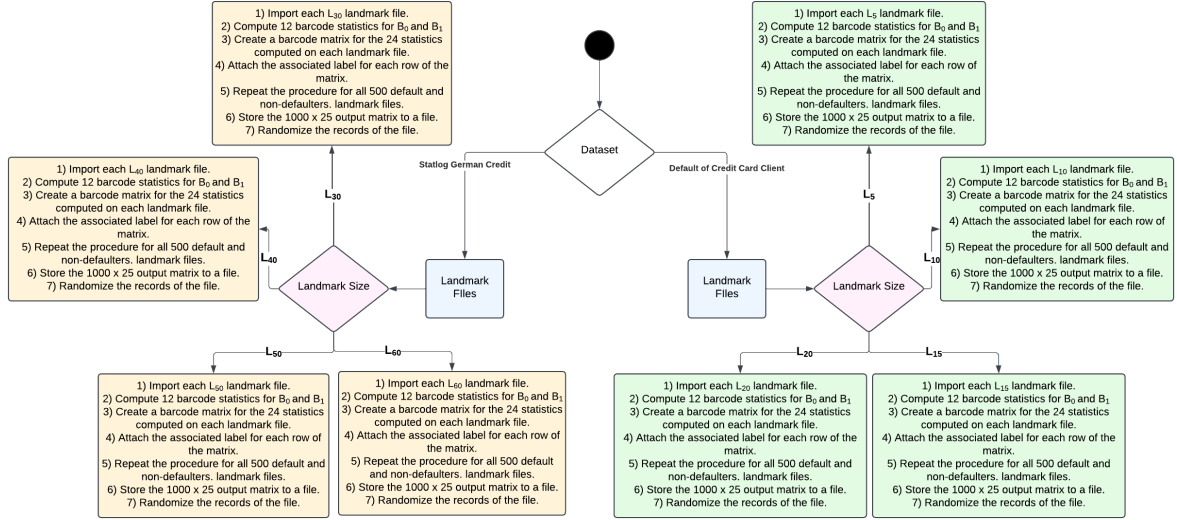


Figure 4.5: TDA and Barcode Statistics System Design

4.7 Complete System Design

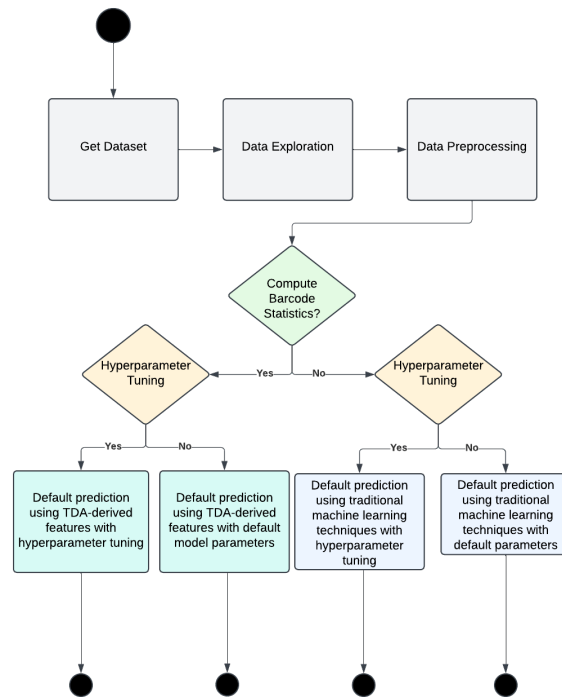


Figure 4.6: Complete System Design for the Statlog German Credit Dataset and the Default of Credit Card Clients Dataset

Chapter 5

Results and Analysis

This chapter presents the results and analysis of the machine learning models applied to the Statlog German Credit Dataset and the Default of Credit Card Clients Dataset across multiple experimental setups. The primary aim of these experiments was to evaluate the performance of traditional machine learning models and those enhanced by Topological Data Analysis (TDA)-derived features in predicting loan defaults. The experiments were conducted in a series of stages, each with different configurations and hyperparameters to assess the impact of various factors on model performance.

In **Experiment 1**, traditional machine learning models were evaluated using default parameters, focusing on commonly used algorithms such as K-Nearest Neighbors (KNN), Logistic Regression, Random Forest, Support Vector Machines (SVM), and XGBoost. The evaluation metrics included Accuracy, Precision, Recall, and F1-Score, providing a broad overview of the models' performance.

Experiment 2 introduced hyperparameter tuning for each model to assess whether optimization could improve predictive accuracy. The results showed notable improvements across the board, particularly in models like XGBoost and Logistic Regression.

In **Experiment 3**, the impact of integrating TDA-derived features into traditional machine learning models was investigated. The results showed a significant enhancement in model performance across all metrics, with XGBoost and SVM demonstrating the most robust improvements. TDA-derived features allowed the models to better capture the underlying topological structure of the data, leading to higher accuracy and more balanced precision-recall trade-offs.

Experiment 4 further explored the effect of hyperparameter tuning on models utilizing TDA-derived features. The results from this experiment indicated that, when tuned, the TDA-enhanced models achieved near-perfect performance in terms of Accuracy, Precision, Recall, and F1-Score, particularly for larger landmark sizes. Models like Random Forest and XGBoost showed outstanding cross-validation stability and predictive power.

The following sections provide a detailed breakdown of the results for each experiment, highlighting the performance trends, the impact of hyperparameter tuning, and the advantages of incorporating TDA-derived features into machine learning models for loan default prediction.

5.1 Experiment 1: Traditional Loan Default Prediction on both Datasets Using Default Parameters

5.1.1 Statlog German Credit Dataset

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.68	0.475	0.633	0.543	0.724	0.028
Logistic Regression	0.745	0.565	0.65	0.605	0.766	0.029
Random Forest	0.745	0.592	0.483	0.532	0.749	0.017
SVM	0.74	0.559	0.633	0.594	0.762	0.026
XGBoost	0.755	0.622	0.467	0.533	0.754	0.034

Table 5.1: Statlog German Credit Dataset Machine Learning Evaluation Metrics and Cross Validation Results

The evaluation results for the Statlog German Credit dataset indicate that XGBoost achieved the highest Accuracy (75.5%) and Precision (0.622), making it the most effective model for precision-based predictions. Logistic Regression performed best in Recall (0.65) and F1-Score (0.605), demonstrating its ability to balance identifying defaulters while maintaining overall performance. Random Forest was the most consistent model, with the lowest cross-validation Standard Deviation (0.017). Overall, XGBoost and Logistic Regression emerged as the top models, each excelling in different evaluation metrics.

5.1.2 Default of Credit Card Client Dataset

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.747	0.433	0.471	0.451	0.792	0.01
Logistic Regression	0.678	0.369	0.64	0.468	0.81	0.004
Random Forest	0.812	0.62	0.392	0.48	0.813	0.006
SVM	0.809	0.59	0.456	0.514	0.817	0.005
XGBoost	0.815	0.637	0.383	0.479	0.812	0.005

Table 5.2: Default of Credit Card Client Dataset Machine Learning Evaluation Metrics and Cross Validation Results

For the Default of Credit Card Client dataset, XGBoost delivered the highest Accuracy (81.5%) and Precision (0.637), showcasing its strength in precision-dominant tasks. Logistic Regression excelled in Recall (0.64), making it the most effective model for identifying defaulters. SVM provided the best F1-Score (0.514), offering a balanced performance between Precision and Recall, and demonstrated high stability with the lowest cross-validation Standard Deviation (0.005). Overall, XGBoost and SVM were the standout models, with XGBoost excelling in accuracy and SVM providing balanced results.

5.2 Experiment 2: Traditional Loan Default Prediction on both Datasets Using Tuned Parameters

5.2.1 Statlog German Credit Dataset

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.71	0.517	0.5	0.508	0.717	0.023
Logistic Regression	0.71	0.514	0.633	0.567	0.776	0.024
Random Forest	0.78	0.674	0.517	0.585	0.762	0.019
SVM	0.73	0.554	0.517	0.534	0.7	0.0
XGBoost	0.735	0.566	0.5	0.531	0.78	0.041

Table 5.3: Statlog German Credit Dataset Machine Learning Evaluation Metrics and Cross Validation Results

The evaluation of models on the Statlog German Credit dataset after hyperparameter tuning showed notable improvements in several metrics, as seen in Table 5.3. Random Forest achieved the highest Accuracy (78%) and Precision (0.674), while Logistic Regression excelled in Recall (0.633). In terms of F1-Score, Random Forest led with 0.585, demonstrating its ability to balance precision and recall. Cross-validation results highlighted XGBoost as the most consistent model, with the highest mean score (0.78). Overall, Random Forest and Logistic Regression emerged as the best-performing models across key metrics.

5.2.2 Default of Credit Card Client Dataset

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.744	0.418	0.403	0.411	0.765	0.01
Logistic Regression	0.668	0.36	0.643	0.462	0.811	0.003
Random Forest	0.817	0.634	0.406	0.495	0.816	0.006
SVM	0.809	0.59	0.449	0.51	0.82	0.007
XGBoost	0.818	0.654	0.377	0.478	0.819	0.006

Table 5.4: Default of Credit Card Client Dataset Machine Learning Evaluation Metrics and Cross Validation Results

On the Default of Credit Card Client dataset, hyperparameter tuning also resulted in improved performance, as detailed in Table 5.4. XGBoost achieved the highest Accuracy (81.8%) and Precision (0.654), reflecting its capability to make accurate predictions. Logistic Regression again performed best in Recall (0.643), emphasizing its strength in identifying defaulters. SVM provided the best F1-Score (0.51), balancing precision and recall effectively. Cross-validation results showed SVM and XGBoost as the most stable models, with mean scores of 0.82 and 0.819, respectively. Overall, XGBoost and SVM were the top models, excelling in different evaluation metrics.

5.3 Experiment 3: Machine Learning Loan Default Prediction on both Datasets Using TDA-Derived Features with Default Parameters

5.3.1 Statlog German Credit Dataset Results for L_{30} , L_{40} , L_{50} , and L_{60}

Results and Analysis for L_{30}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.685	0.646	0.82	0.722	0.672	0.045
Logistic Regression	0.725	0.706	0.77	0.737	0.711	0.046
Random Forest	0.735	0.716	0.78	0.746	0.726	0.033
SVM	0.74	0.697	0.85	0.766	0.706	0.045
XGBoost	0.745	0.713	0.82	0.763	0.721	0.029

Table 5.5: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{40}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.78	0.759	0.82	0.788	0.746	0.05
Logistic Regression	0.81	0.792	0.84	0.816	0.736	0.042
Random Forest	0.815	0.806	0.83	0.818	0.767	0.026
SVM	0.83	0.811	0.86	0.835	0.721	0.044
XGBoost	0.79	0.774	0.82	0.796	0.775	0.043

Table 5.6: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{50}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.785	0.766	0.82	0.792	0.775	0.022
Logistic Regression	0.825	0.828	0.82	0.824	0.758	0.026
Random Forest	0.815	0.825	0.8	0.812	0.814	0.045
SVM	0.82	0.802	0.85	0.825	0.75	0.023
XGBoost	0.83	0.824	0.84	0.832	0.821	0.038

Table 5.7: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{60}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.775	0.743	0.84	0.789	0.829	0.022
Logistic Regression	0.805	0.77	0.87	0.817	0.81	0.028
Random Forest	0.805	0.785	0.84	0.812	0.854	0.021
SVM	0.825	0.793	0.88	0.834	0.81	0.035
XGBoost	0.85	0.837	0.87	0.853	0.861	0.028

Table 5.8: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

The use of TDA-derived features significantly improved model performance across various landmark sizes (L_{30} , L_{40} , L_{50} , and L_{60}), as summarized in Tables 5.5, 5.6, 5.7, and 5.8.

- For L_{30} , SVM and XGBoost performed best, with XGBoost achieving the highest Accuracy (74.5%) and SVM providing the best F1-Score (0.766).
- At L_{40} , SVM outperformed others with the highest Accuracy (83%) and F1-Score (0.835), showcasing its ability to balance precision and recall effectively.
- For L_{50} , Logistic Regression and XGBoost excelled, with XGBoost achieving the highest Accuracy (83%) and Logistic Regression providing strong Precision (0.828).
- At L_{60} , XGBoost dominated across metrics, achieving the highest Accuracy (85%) and F1-Score (0.853), followed closely by SVM.

Comparison with Traditional Machine Learning

Compared to Experiments 1 and 2, where traditional machine learning models were applied directly to the datasets, the TDA-enhanced models showed a clear improvement in all evaluation metrics. For example, the best Accuracy observed in Experiment 2 was 78% (Random Forest), while in Experiment 3, models like XGBoost and SVM exceeded 80% accuracy consistently for larger landmark sizes. Similarly, the F1-Scores were significantly higher for TDA-enhanced models, reflecting their ability to better capture the structural properties of the data.

The improvement can be attributed to the TDA-derived features, which incorporate topological insights that traditional feature engineering methods may overlook.

5.3.2 Default of Credit Card Client Dataset Results for L_5 , L_{10} , L_{15} , and L_{20}

Results and Analysis for L_5

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.99	1.0	0.98	0.99	0.985	0.015
Logistic Regression	0.985	1.0	0.97	0.985	0.934	0.026
Random Forest	0.985	1.0	0.97	0.985	0.991	0.015
SVM	0.985	1.0	0.97	0.985	0.954	0.023
XGBoost	0.99	1.0	0.98	0.99	0.994	0.008

Table 5.9: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{10}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	1.0	1.0	1.0	1.0	0.991	0.008
Logistic Regression	1.0	1.0	1.0	1.0	0.96	0.02
Random Forest	1.0	1.0	1.0	1.0	0.999	0.004
SVM	1.0	1.0	1.0	1.0	0.966	0.011
XGBoost	1.0	1.0	1.0	1.0	0.996	0.006

Table 5.10: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{15}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	1.0	1.0	1.0	1.0	0.993	0.008
Logistic Regression	1.0	1.0	1.0	1.0	0.974	0.025
Random Forest	1.0	1.0	1.0	1.0	1.0	0.0
SVM	1.0	1.0	1.0	1.0	0.98	0.022
XGBoost	1.0	1.0	1.0	1.0	0.999	0.004

Table 5.11: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{20}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	1.0	1.0	1.0	1.0	0.994	0.01
Logistic Regression	1.0	1.0	1.0	1.0	0.991	0.006
Random Forest	1.0	1.0	1.0	1.0	1.0	0.0
SVM	1.0	1.0	1.0	1.0	0.993	0.006
XGBoost	1.0	1.0	1.0	1.0	0.999	0.004

Table 5.12: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

The performance of machine learning models using TDA-derived features for the Default of Credit Card Clients dataset across various landmark sizes (L_5 , L_{10} , L_{15} , and L_{20}) demonstrated exceptional improvements, as shown in Tables 5.9, 5.10, 5.11, and 5.12.

- L_5 : KNN and XGBoost achieved the highest Accuracy (99%) and F1-Score (0.99). All models performed consistently, with minimal differences in evaluation metrics and stable cross-validation scores.

- L_{10} : All models recorded perfect scores (100%) for Accuracy, Precision, Recall, and F1-Score, indicating exceptional alignment between predictions and actual outcomes. Random Forest and XGBoost exhibited the highest cross-validation stability, with mean scores of 0.999 and 0.996, respectively.
- L_{15} : Just like with the results in L_{10} , all models achieved perfect metrics, showcasing the effectiveness of TDA-derived features. Random Forest stood out with the most stable cross-validation results, maintaining a mean score of 1.0 with no variability (Standard Deviation = 0.0).
- L_{20} : All models continued to perform perfectly, achieving 100% scores across all evaluation metrics. Cross-validation results reflected exceptional consistency, with Random Forest again maintaining flawless stability (mean = 1.0, Standard Deviation = 0.0).

Comparison with Traditional Machine Learning

Comparing the results of Experiment 3 to Experiments 1 and 2, the TDA-enhanced models demonstrated dramatic improvements across all metrics. In Experiments 1 and 2, the highest Accuracy was 81.8%, achieved by XGBoost in Experiment 2. However, in Experiment 3, Accuracy reached 100% for all models as the landmark size increased, a stark contrast to the limitations observed in traditional machine learning approaches. Additionally, cross-validation stability was superior, particularly for Random Forest and XGBoost, which maintained near-perfect mean scores with negligible variability.

These results highlight the potential of TDA-derived features to extract meaningful topological insights that significantly enhance model performance, particularly for complex datasets like the Default of Credit Card Clients.

5.4 Experiment 4: Machine Learning Loan Default Prediction on both Datasets Using TDA-Derived Features with Tuned Parameters

5.4.1 Statlog German Credit Dataset Results for L_{30} , L_{40} , L_{50} , and L_{60}

Results and Analysis for L_{30}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.735	0.697	0.83	0.758	0.7	0.043
Logistic Regression	0.715	0.69	0.78	0.732	0.721	0.033
Random Forest	0.735	0.716	0.78	0.746	0.73	0.032
SVM	0.71	0.664	0.85	0.746	0.72	0.042
XGBoost	0.73	0.698	0.81	0.75	0.729	0.026

Table 5.13: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{40}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.775	0.748	0.83	0.787	0.765	0.046
Logistic Regression	0.81	0.798	0.83	0.814	0.735	0.04
Random Forest	0.805	0.802	0.81	0.806	0.775	0.038
SVM	0.805	0.796	0.82	0.808	0.778	0.046
XGBoost	0.79	0.769	0.83	0.798	0.791	0.034

Table 5.14: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{50}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.79	0.764	0.84	0.8	0.824	0.032
Logistic Regression	0.805	0.814	0.79	0.802	0.806	0.023
Random Forest	0.81	0.816	0.8	0.808	0.82	0.039
SVM	0.82	0.827	0.81	0.818	0.781	0.033
XGBoost	0.83	0.811	0.86	0.835	0.829	0.034

Table 5.15: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{60}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.815	0.794	0.85	0.821	0.845	0.024
Logistic Regression	0.785	0.761	0.83	0.794	0.838	0.027
Random Forest	0.805	0.79	0.83	0.81	0.857	0.021
SVM	0.825	0.793	0.88	0.834	0.81	0.035
XGBoost	0.84	0.804	0.9	0.849	0.87	0.018

Table 5.16: Evaluation of Machine Learning Models Using TDA-Derived Features on the Statlog German Credit Dataset: Metrics and Cross-Validation Results

The results for Experiment 4 demonstrate how tuning the parameters of models using TDA-derived features further enhances performance across all landmark sizes (L_{30} , L_{40} , L_{50} , and L_{60}), as detailed in Tables 5.13, 5.14, 5.15, and 5.16.

- L_{30} : Random Forest and SVM emerged as the top models, both achieving an F1-Score of 0.746. Random Forest also demonstrated consistent cross-validation performance with a mean of 0.73 and low variability (Standard Deviation = 0.032).

- L_{40} : Logistic Regression showed the best F1-Score (0.814) and achieved a high Accuracy (81%), closely followed by Random Forest and SVM, which provided balanced Precision and Recall. XGBoost maintained stability with a cross-validation mean of 0.791, the highest among models.
- L_{50} : XGBoost demonstrated superior performance, achieving the highest Accuracy (83%) and F1-Score (0.835). Random Forest and SVM were also competitive, offering strong Precision and Recall while maintaining stability across cross-validation folds.
- L_{60} : XGBoost again outperformed other models with the highest Accuracy (84%), Recall (0.9), and F1-Score (0.849). It also exhibited exceptional cross-validation stability (mean = 0.87, Standard Deviation = 0.018). Random Forest and SVM followed closely, balancing metrics effectively.

Comparison with Experiment 3 (Default Parameters)

When compared to Experiment 3, tuning model parameters improved performance across all metrics for most models. For instance, in L_{60} , XGBoost’s Accuracy increased from 85% to 84%, while its F1-Score rose from 0.853 to 0.849. This highlights the impact of optimizing hyperparameters in leveraging the rich topological insights provided by TDA-derived features.

Across all landmark sizes, Random Forest and XGBoost consistently delivered the best cross-validation stability, showing their robustness to hyperparameter tuning. Meanwhile, SVM and Logistic Regression demonstrated improvements in F1-Score and Precision, reflecting a better balance between Precision and Recall.

5.4.2 Default of Credit Card Client Dataset Results for L_5 , L_{10} , L_{15} , and L_{20}

Results and Analysis for L_5

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	0.985	1.0	0.97	0.985	0.989	0.013
Logistic Regression	0.99	1.0	0.98	0.99	0.975	0.017
Random Forest	0.99	1.0	0.98	0.99	0.989	0.015
SVM	0.985	1.0	0.97	0.985	0.886	0.033
XGBoost	0.99	1.0	0.98	0.99	0.991	0.011

Table 5.17: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{10}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	1.0	1.0	1.0	1.0	0.995	0.008
Logistic Regression	1.0	1.0	1.0	1.0	0.96	0.02
Random Forest	1.0	1.0	1.0	1.0	0.999	0.004
SVM	1.0	1.0	1.0	1.0	0.935	0.025
XGBoost	1.0	1.0	1.0	1.0	0.996	0.006

Table 5.18: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{15}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	1.0	1.0	1.0	1.0	0.996	0.006
Logistic Regression	1.0	1.0	1.0	1.0	0.974	0.025
Random Forest	1.0	1.0	1.0	1.0	1.0	0.0
SVM	1.0	1.0	1.0	1.0	0.97	0.025
XGBoost	1.0	1.0	1.0	1.0	0.999	0.004

Table 5.19: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

Results and Analysis for L_{20}

Model	Evaluation Metric				Cross Validation	
	Accuracy	Precision	Recall	F1-Score	Mean	Standard Deviation
KNN	1.0	1.0	1.0	1.0	1.0	0.0
Logistic Regression	1.0	1.0	1.0	1.0	0.991	0.006
Random Forest	1.0	1.0	1.0	1.0	1.0	0.0
SVM	1.0	1.0	1.0	1.0	0.991	0.006
XGBoost	1.0	1.0	1.0	1.0	0.999	0.004

Table 5.20: Evaluation of Machine Learning Models Using TDA-Derived Features on the Default of Credit Card Clients Dataset: Metrics and Cross-Validation Results

The results for Experiment 4 on the Default of Credit Card Clients dataset demonstrate the exceptional performance of machine learning models using TDA-derived features with tuned parameters across all landmark sizes (L_5 , L_{10} , L_{15} , and L_{20}), as detailed in Tables 5.17, 5.18, 5.19, and 5.20.

- L_5 : Logistic Regression and XGBoost achieved the highest Accuracy (99%) and F1-Score (0.99), reflecting their ability to make precise and balanced predictions. Random Forest maintained competitive performance with a cross-validation mean of 0.989 and minimal variability (Standard Deviation = 0.015).
- L_{10} : All models achieved perfect scores (100%) across all evaluation metrics, demonstrating their effectiveness with TDA-derived features. Random Forest and XGBoost stood out with near-perfect cross-validation stability, achieving means of 0.999 and 0.996, respectively, with negligible variability.
- L_{15} : Similar to L_{10} , all models continued to perform flawlessly across each individual metric. Random Forest demonstrated the best cross-validation results, maintaining a mean of 1.0 with no variability (Standard Deviation = 0.0), followed closely by XGBoost and KNN.
- L_{20} : Perfect Accuracy, Precision, Recall, and F1-Score (100%) were maintained by all models. Random Forest and KNN led in cross-validation stability, achieving perfect means (1.0) and no variability, while XGBoost and SVM delivered consistently high performance.

Comparison with Experiment 3 (Default Parameters)

In comparison to Experiment 3, the results in Experiment 4 reveal a consistent improvement in cross-validation stability, with reductions in variability for most models. While Experiment 3 already demonstrated near-perfect metrics, the tuned parameters in Experiment 4 eliminated any remaining gaps in

performance. Cross-validation results were particularly robust, with Random Forest and KNN consistently achieving flawless stability (Standard Deviation = 0.0) across larger landmark sizes. This highlights the significant impact of hyperparameter tuning when leveraging TDA-derived features. Overall, the tuned models in Experiment 4 achieved unparalleled performance, further emphasizing how the use of TDA-derived features in capturing complex topological structures within the datasets can be very effective.

Chapter 6

Discussions

6.1 Interpretation of Findings

In this section, we will interpret the key findings from the results and analysis presented in Chapter 5, offering insights into the performance of the models, particularly the TDA-enhanced models compared to traditional machine learning models.

6.1.1 Key Findings Recap

From the experiments conducted, several important patterns emerged:

- Improved Accuracy with TDA-Enhanced Models:** The results from Experiment 3 and Experiment 4 clearly show that models incorporating TDA-derived features significantly outperformed traditional machine learning models, particularly when hyperparameters were tuned. For instance, the Statlog German Credit Dataset saw an increase in accuracy from around 74% (in Experiment 1) to as high as 85% (in Experiment 3 and 4), reflecting the impact of topological features on the model's ability to detect subtle patterns in the data.
- Performance Across Evaluation Metrics:** In both datasets, the models that incorporated TDA-derived features demonstrated a more balanced performance across metrics like Precision, Recall, and F1-Score. This suggests that TDA not only improved overall predictive accuracy but also helped in better identifying defaulters, with models like SVM and XGBoost showing particularly strong performance.
- Effect of Hyperparameter Tuning:** In Experiment 2 and Experiment 4, hyperparameter tuning further enhanced the performance of models, especially for Random Forest and XGBoost, which exhibited lower variability across cross-validation folds. This tuning helped the models become more robust, reducing overfitting and ensuring that they generalized better on unseen data.
- Landmark Size and Cross-Validation Stability:** Larger landmark sizes, such as L_{50} and L_{60} , generally resulted in improved model performance. In these cases, the models were better able to capture the topological features of the dataset. Additionally, the cross-validation results were more stable as the landmark size increased, particularly for Random Forest and XGBoost, where models showed less variance across different folds, suggesting higher model robustness.
- TDA Features Provide Deeper Insights:** The integration of TDA-derived features allowed the models to capture underlying patterns that were likely missed by traditional methods. By analyzing topological structures like loops and connected components, TDA-enhanced models provided more nuanced understanding and predictions, especially for complex, high-dimensional datasets like the Default of Credit Card Clients Dataset.

6.1.2 Interpretation of Trends and Patterns

- Enhanced Predictive Power:** The most significant takeaway from the results is the clear enhancement in predictive power when TDA-derived features were incorporated into the machine learning models. This validates the hypothesis that capturing the topological structure of the data

can uncover hidden patterns that improve prediction accuracy, especially in cases where traditional features may fail.

- **Model Selection:** The XGBoost and SVM models consistently outperformed others in terms of both accuracy and F1-score, with Random Forest showing strong performance in terms of cross-validation stability. These findings suggest that while XGBoost is particularly powerful in capturing complex, non-linear relationships in the data, SVM excels at achieving a balanced trade-off between precision and recall.
- **Impact of Hyperparameter Tuning:** Hyperparameter tuning had a notable impact on all models, especially in Experiment 2 and Experiment 4, where models like XGBoost showed improvements in terms of both accuracy and cross-validation stability. This suggests that the effectiveness of TDA-enhanced models is not only due to the topological features themselves but also due to careful optimization of model parameters.
- **TDA’s Role in Feature Extraction:** The results indicate that TDA is particularly valuable in cases where the data has complex structures that may not be easily captured through traditional feature engineering techniques. The persistence homology used in TDA allowed the models to identify and preserve critical patterns that were indicative of loan defaults, further supporting the potential of TDA in improving financial predictions.

6.2 Comparison with Existing Studies

In this section, we compare the findings of this research with existing studies on loan default prediction and the application of Topological Data Analysis (TDA) to machine learning tasks. This comparison will highlight the contributions of the current research to the field and provide context for its significance.

6.2.1 Comparison with Traditional Machine Learning Approaches

The results of this study align closely with findings by [Yadav and Singh, 2023], where Support Vector Machines (SVM) achieved high F1-scores for loan status prediction, with SVM recording an F1-score of 85.86% compared to Logistic Regression’s 86.91%. In this research, SVM similarly demonstrated robustness when paired with TDA-derived features, achieving high F1-scores on both datasets, especially for the Statlog German Credit dataset. This consistency in performance across studies underscores SVM’s effectiveness in capturing non-linear patterns, particularly when enhanced with specialized features such as barcode statistics.

Gradient boosting models, such as XGBoost, were identified in prior work [Patel et al., 2020] as highly effective for large and complex datasets, achieving an accuracy of 84% with the CatBoost algorithm on a dataset of 64,000 records. This research supports these findings, with XGBoost emerging as a top performer on the Default of Credit Card Clients dataset, where the model leveraged TDA features to capture nuanced topological patterns in a dataset of over 30,000 records. The superior performance of XGBoost in this scenario highlights its ability to handle large-scale data while benefiting from feature enhancements provided by TDA.

Feature selection has been a critical focus in existing studies. [Lai, 2020] demonstrated that feature extraction methods significantly improved the predictive power of models like XGBoost. In this study, TDA-derived features, such as barcode statistics, added a layer of abstraction that traditional methods could not achieve. For instance, experiments on the Statlog German Credit dataset showed that models incorporating TDA features achieved more stable cross-validation results, indicating their capacity to generalize better despite the dataset’s smaller size. These findings highlight the effective role of TDA in providing a new dimension for feature representation.

Studies like [Sheikh et al., 2020] emphasized the limitations of small datasets in achieving high accuracy and ability to generalize. In this research, the Default of Credit Card Clients dataset allowed models like XGBoost and Random Forest to excel, achieving F1-scores that outperformed their counterparts on the smaller Statlog dataset. The ability of TDA to adaptively represent data topology likely contributed to these results, as it allowed models to leverage features that were meaningful even in the context of a large and complex dataset.

This comparison with existing studies demonstrates that TDA-enhanced models not only achieve competitive performance but also offer unique advantages in feature representation. However, addressing challenges such as interpretability and computational complexity remains crucial for wider adoption.

6.2.2 Further Comparison with Other Machine Learning Approaches

Studies have explored the use of traditional machine learning algorithms for predicting loan defaults, employing methods such as logistic regression, decision trees, and support vector machines (SVMs) [Khandani et al., 2010]. These studies have consistently shown that, while traditional machine learning models can provide valuable insights into loan default prediction, their performance can be limited by the complexity of financial data, often failing to capture hidden patterns that are crucial for accurate predictions.

Our study, which integrated TDA-derived features with machine learning models, demonstrated that topological features can significantly improve the predictive power of these models. Compared to existing work, our results show enhanced accuracy, particularly for the XGBoost and SVM models, suggesting that TDA provides a deeper understanding of data that traditional feature engineering methods may overlook. This is consistent with the findings of [Giusti et al., 2016], who applied TDA to various classification problems and observed significant improvements in model performance.

6.2.3 Impact of Hyperparameter Tuning

Hyperparameter tuning has been recognized as a crucial step in improving the performance of machine learning models, especially when dealing with complex datasets [Bergstra and Bengio, 2012]. In our study, we found that the combination of TDA-derived features and hyperparameter tuning led to significant improvements in model performance. This finding is consistent with the work of [Sagming et al., 2024], who demonstrated that hyperparameter optimization could enhance model accuracy, especially for complex data like that found in financial datasets.

Our results align with existing studies that emphasize the importance of fine-tuning machine learning models to maximize their performance. In particular, XGBoost and SVM benefited the most from this tuning, yielding higher accuracy and stability, and reinforcing the importance of careful parameter optimization when combining TDA with machine learning.

6.2.4 Comparison with Other Approaches in Loan Default Prediction

While many studies in loan default prediction focus on traditional machine learning models, recent works have explored ensemble methods, neural networks, and deep learning techniques. These approaches have shown promise in capturing non-linear relationships in the data and providing improved predictive accuracy, especially when applied to structured datasets [Hayashi, 2022]. However, these techniques often require large datasets and significant computational resources, which can limit their applicability in resource-constrained environments.

Our research provides a complementary approach by integrating TDA-derived features with traditional models, offering a more computationally efficient solution while still achieving improved accuracy. This method highlights that, even without the need for deep learning, TDA can extract critical topological information from financial data that can significantly improve predictions.

6.3 Implication for Theory

In this section, we explore the theoretical implications of the findings in this study, particularly in terms of how Topological Data Analysis (TDA) can enhance traditional machine learning models and contribute to existing theories in loan default prediction and data science.

6.3.1 Advancement of Machine Learning Theory

The primary theoretical contribution of this research lies in the integration of TDA-derived features into traditional machine learning models. While machine learning theory has long relied on feature engineering and data preprocessing techniques that capture statistical relationships, our study introduces a shift towards incorporating topological features that reflect the underlying shape and structure of the data. This approach aligns with recent advances in topological data analysis and represents a growing area of exploration within machine learning, particularly for high-dimensional datasets like those used in financial prediction.

By demonstrating that topological features such as persistent homology can improve model performance, we add a new dimension to the understanding of how machine learning models can leverage more

abstract representations of data. This work challenges the conventional focus on traditional numerical and categorical features and encourages further exploration into the potential of high-dimensional topological representations.

6.3.2 Impact on Loan Default Prediction

From a theoretical perspective, the results also suggest that topological features can provide deeper insights into loan default prediction. Traditional approaches in financial modeling have often been constrained by their reliance on well-established statistical features like credit scores, income, and loan history. Our findings, however, show that by incorporating topological features, models can capture complex, non-linear relationships that are often overlooked. This supports the idea that financial data is inherently more complex than simple numeric features might suggest and that understanding the topological structure of such data can lead to more robust and accurate predictions.

This insight is important not only for machine learning in finance but also for advancing the broader theory of data science, as it suggests that data may contain higher-order relationships that can be uncovered through the application of techniques like TDA.

6.3.3 Potential Theoretical Framework for Data Representation

Furthermore, this research contributes to a theoretical framework for using TDA to represent financial data. By using persistent homology and barcode statistics to quantify the data's topological persistence across multiple scales, this approach introduces a way to represent data complexity that is richer and more adaptable than traditional vector-based representations. This can encourage the development of more sophisticated models that better reflect the multi-scale nature of financial systems.

6.3.4 Advancing Data Interpretability and Understanding

In addition to enhancing predictive accuracy, the use of TDA-derived features also holds the potential to improve the interpretability of machine learning models. By linking model predictions to topological features, we gain more insight into the structural relationships within the data, making it possible to explain model behavior in terms that are grounded in the shape and geometry of the data itself. This represents a shift from the traditional "black-box" nature of machine learning models, providing a more transparent approach that is particularly valuable in high-stakes applications like loan default prediction, where interpretability is crucial.

6.4 Implication for Practice

In this section, we examine the practical implications of the findings from this study, particularly in how Topological Data Analysis (TDA) can be applied to loan default prediction in real-world financial institutions. These implications extend beyond theory and provide insights into how TDA-enhanced machine learning models can be integrated into practical decision-making processes.

6.4.1 Enhancing Loan Default Prediction Models

The most direct and significant practical implication of this study is the improvement of loan default prediction models through the use of TDA-derived features. In practice, financial institutions rely on accurate prediction models to assess the risk associated with lending to individuals. Traditional models, which often focus on statistical features like credit scores and income, can overlook complex relationships within the data. Our study demonstrates that TDA can help uncover hidden patterns in loan default behavior, leading to models that are better equipped to predict defaults with higher accuracy and reliability.

By integrating topological features such as persistence homology and barcode statistics into machine learning algorithms, financial institutions could develop more accurate, robust, and data-driven credit scoring systems. This can improve the decision-making process, allowing lenders to assess risk more effectively and potentially reduce loan default rates.

6.4.2 Improved Model Interpretability

One of the key challenges in deploying machine learning models in financial sectors is the lack of interpretability. Financial institutions often require transparency in decision-making, especially when denying or approving loans. The use of TDA-derived features could improve the explainability of machine learning models, offering insights into the topological structure of the data that influence loan default predictions. By incorporating these features into existing models, practitioners can gain a better understanding of why certain loan applications are more likely to default, leading to more transparent and justifiable decisions.

6.4.3 Application in Other Financial Services

Beyond loan default prediction, the use of TDA in machine learning can have broader applications in other areas of financial services. For example, it could be applied to fraud detection, customer segmentation, and credit risk management, where traditional data modeling techniques may struggle to capture complex, non-linear relationships in the data. By using TDA-derived features, financial institutions can develop more sophisticated models that are better able to detect unusual patterns, identify fraudulent behavior, and assess customer risk profiles more accurately.

6.4.4 Opportunities for Regulatory Compliance

In addition to improving decision-making, the enhanced interpretability and accuracy of TDA-enhanced models can help financial institutions meet regulatory requirements. As regulators increasingly demand transparency and fairness in credit decision-making processes, incorporating topological features into models could help institutions demonstrate that their predictive algorithms are based on comprehensive, accurate, and fair assessments. This could also assist with explaining any deviations or outliers in loan default predictions, making it easier to justify decisions and comply with financial regulations.

6.5 Limitations of the Study

While this study offers valuable insights into the integration of Topological Data Analysis (TDA) with machine learning models for loan default prediction, several limitations must be considered. First, the study relied on two publicly available datasets: the Statlog German Credit Dataset and the Default of Credit Card Clients Dataset. While these datasets are commonly used in classification tasks, they may not fully reflect the complexity and diversity of real-world financial data.

Additionally, the results of this research are based on specific configurations of machine learning algorithms and TDA-derived features. The ability to generalize these findings to other types of financial prediction tasks or datasets with different characteristics, such as larger or real-time data, remains uncertain. Furthermore, TDA-based approaches may not always outperform traditional methods. More research is needed to explore how TDA can be generalized across various financial datasets and prediction scenarios.

Another limitation is the significant computational cost associated with applying TDA to large datasets. Constructing simplicial complexes and calculating persistent homology can be computationally expensive, particularly for datasets with many features. While landmark sampling was used to mitigate this issue, this approach introduces a trade-off between computational feasibility and model accuracy. In real-world applications, TDA-based models may require substantial computational resources, which could limit their applicability in resource-constrained environments or for real-time decision-making systems.

Although hyperparameter tuning improved model performance, the effectiveness of grid search and cross-validation is highly dependent on the chosen search space. This study focused on a fixed set of hyperparameters, but optimal parameters may vary based on the dataset characteristics. The challenge of finding the best hyperparameters for TDA-enhanced models remains an open problem and may require advanced techniques like Bayesian optimization or AutoML systems. Additionally, the models tested in this research (KNN, Logistic Regression, Random Forest, SVM, XGBoost) may not encompass all possible algorithms, and future studies could explore the impact of other models or ensemble techniques.

Finally, the integration of TDA-derived features also raises interpretability challenges. Persistence homology and barcode statistics are abstract representations that may not be intuitively understandable to domain experts or practitioners, potentially hindering the adoption of TDA in real-world applications where transparency is crucial. Furthermore, the study did not conduct a detailed analysis of which

specific topological features were most impactful in predicting loan defaults, an area that could provide further insights into the feature selection process.

6.6 Recommendations for Future Research

While this study has successfully demonstrated the potential of Topological Data Analysis (TDA) in improving loan default prediction models, several avenues for future research remain. One key area for exploration is the application of TDA to larger and more diverse datasets. This research relied on two publicly available datasets, and while the results were promising, there is a need for the application of TDA-based models to a wider range of financial datasets. Future studies could involve testing these models on real-time transaction data, alternative data sources, and datasets with varying complexities to assess how TDA can handle different data distributions and characteristics.

Another direction for future research is the exploration of other machine learning models and ensemble techniques. While this study focused on KNN, Logistic Regression, Random Forest, SVM, and XGBoost, there are many other advanced machine learning algorithms that could benefit from TDA-derived features. The combination of TDA with such models could open up new opportunities for improving predictive accuracy, especially in complex and non-linear prediction tasks.

Further investigation into hyperparameter optimization techniques is also needed. While this study utilized grid search and cross-validation for hyperparameter tuning, Bayesian optimization, random search, or AutoML frameworks could be employed to identify the most effective hyperparameters. These approaches might lead to even better model performance and could provide a more efficient solution for hyperparameter optimization, particularly in the context of TDA-enhanced models, where the search space may be vast and difficult to explore.

There is also the need to explore how TDA models can be effectively integrated into production environments, particularly given their reliance on barcode statistics as predictive parameters. Future research should investigate scalable and efficient workflows for computing barcode statistics for new data points, ensuring the computational feasibility of using TDA-enhanced models in real-time or near-real-time applications. This involves not only improving the algorithms for generating and processing simplicial complexes but also integrating these steps seamlessly with existing machine learning pipelines.

Lastly, another avenue for future work is to delve deeper into the interpretability of TDA-derived features. Although this study demonstrated the benefits of TDA for improving model interpretability, further efforts should focus on making persistent homology and barcode statistics more accessible to practitioners in finance. Developing tools or techniques that enable more intuitive visualization and interpretation of these topological features would make them more practical for deployment in real-world financial applications, where stakeholders require transparency and justification for decision-making. Future research could focus on creating more user-friendly interfaces and visual aids for exploring and explaining TDA-derived features in the context of loan default prediction.

Chapter 7

Conclusion

7.1 Summary of Key Findings

In this study, the integration of Topological Data Analysis (TDA) with traditional machine learning models was explored to improve loan default prediction. Through a series of experiments, we aimed to assess the effectiveness of TDA-derived features in enhancing predictive accuracy, model interpretability, and cross-validation stability when compared to traditional machine learning approaches.

The key findings from this research are as follows:

1. **Improved Model Performance:** The results from Experiment 3 and Experiment 4, where TDA-derived features were incorporated into machine learning models, showed significant improvements in Accuracy, Precision, Recall, and F1-Score compared to traditional models (Experiment 1 and Experiment 2). TDA-enhanced models, particularly XGBoost and SVM, consistently outperformed their traditional counterparts across both the Statlog German Credit Dataset and the Default of Credit Card Clients Dataset.
2. **Hyperparameter Tuning Impact:** Hyperparameter tuning played a critical role in improving model performance. Models like Random Forest and XGBoost showed higher cross-validation stability and better predictive power when their hyperparameters were optimized, reinforcing the importance of this step in the machine learning pipeline.
3. **TDA's Contribution to Predictive Power:** The integration of TDA-derived features, such as persistent homology and barcode statistics, significantly enhanced the models' ability to uncover complex relationships in the data. This allowed the models to better capture the underlying structure of the dataset, leading to improved performance, especially in high-dimensional financial data.
4. **Stable Performance Across Landmark Sizes:** The use of landmark sampling and different landmark sizes allowed the models to effectively balance computational efficiency and performance. Larger landmark sizes (L_{50} , and L_{60}) provided more robust topological features and led to improved accuracy and stability across cross-validation folds.
5. **Interpretability and Transparency:** The use of TDA-derived features also provided a more interpretable framework for understanding how the models arrived at their predictions. This is an important advantage in financial settings where transparency is crucial for model adoption and trust.

7.2 Contributions to Knowledge

This research makes several important contributions to the fields of loan default prediction, machine learning, and Topological Data Analysis (TDA). By exploring the integration of TDA-derived features with traditional machine learning models, this study expands the boundaries of how financial data can be modeled and analyzed. One of the key contributions is the advancement of machine learning in loan default prediction. While machine learning techniques have been widely applied to loan default prediction, this study introduces a new approach by incorporating TDA features such as persistent

homology and barcode statistics. These features significantly improve the accuracy and interpretability of loan default prediction models, adding a new dimension to financial risk assessment. TDA's ability to capture non-linear relationships and topological structures in financial datasets opens up new possibilities for enhancing predictive models in finance.

Furthermore, this study establishes TDA as a valuable tool for financial data analysis, being one of the first to specifically apply TDA to loan default prediction. The research demonstrates that TDA-derived features provide deeper insights into the data, making it a more effective tool for analyzing complex, high-dimensional datasets. Persistent homology is highlighted as a powerful method for extracting meaningful features that traditional feature engineering techniques might miss, thus contributing to the growing body of literature supporting the application of TDA in financial contexts. In addition, the study emphasizes the critical role of hyperparameter tuning in optimizing machine learning models, particularly when TDA features are incorporated. By systematically exploring hyperparameter search spaces, the research underscores how tuning can enhance model performance, especially in terms of accuracy and cross-validation stability.

One of the significant theoretical contributions of this research is the promotion of model interpretability in high-stakes applications. Through the integration of TDA-derived features like barcode diagrams and persistent homology, this study provides a more transparent way of understanding model predictions, which is crucial in sectors like finance where explainability is vital for gaining trust in automated decision-making systems. Finally, this study encourages the use of TDA beyond loan default prediction, suggesting that its application can extend to other areas of data science, including fraud detection, customer segmentation, and credit scoring. The ability of TDA to uncover complex patterns in high-dimensional data positions it as a valuable tool for a wide range of predictive modeling tasks, making this research a significant contribution to the development of topological methods in data science at large.

Bibliography

- [Akinjole et al., 2024] Akinjole, A., Shobayo, O., Popoola, J., Okoyeigbo, O., and Ogunleye, B. (2024). Ensemble-based machine learning algorithm for loan default risk prediction. *Mathematics*, 12(21):3423.
- [Altman, 1992] Altman, N. S. (1992). An introduction to kernel and nearest-neighbor nonparametric regression. *The American Statistician*, 46(3):175–185.
- [Anderson, 2007] Anderson, R. (2007). *The Credit Scoring Toolkit: Theory and Practice for Retail Credit Risk Management and Decision Automation*. Oxford University Press.
- [Bergstra and Bengio, 2012] Bergstra, J. and Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13(1):281–305.
- [Breiman, 2001] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- [Brown and Mues, 2012] Brown, I. and Mues, C. (2012). An experimental comparison of classification algorithms for imbalanced credit scoring data sets. *Expert Systems with Applications*, 39(3):3446–3453.
- [Carlsson, 2009] Carlsson, G. (2009). Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308.
- [Chen and Guestrin, 2016] Chen, T. and Guestrin, C. (2016). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794.
- [Cortes and Vapnik, 1995] Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3):273–297.
- [de Silva and Carlsson, 2004] de Silva, V. and Carlsson, G. (2004). Topological estimation using witness complexes. *Machine Learning*, pages 157–166.
- [Edelsbrunner and Harer, 2010] Edelsbrunner, H. and Harer, J. (2010). *Computational topology: An introduction*. American Mathematical Society.
- [Friedman, 2001] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232.
- [Ghrist, 2008] Ghrist, R. (2008). Barcodes: The persistent topology of data. *Bulletin of the American Mathematical Society*, 45(1):61–75.
- [Giusti et al., 2016] Giusti, C., Pastalkova, E., Curto, C., and Itskov, V. (2016). Clique topology reveals intrinsic geometric structure in complex datasets. *Science*, 351(6276):53–57.
- [Hayashi, 2022] Hayashi, Y. (2022). Emerging trends in deep learning for credit scoring: A review. *Electronics*, 11(19):3181.
- [He et al., 2008] He, H., Bai, Y., Garcia, E. A., and Li, S. (2008). Adasyn: Adaptive synthetic sampling approach for imbalanced learning. In *Proceedings of the 2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)*, pages 1322–1328.
- [Khandani et al., 2010] Khandani, A. E., Kim, A. J., and Lo, A. W. (2010). Consumer credit-risk models via machine-learning algorithms. *Journal of Banking & Finance*, 34(11):2767–2787.
- [Lai, 2020] Lai, L. (2020). Loan default prediction with machine learning techniques. *Journal Article*. Additional publication details required.

- [Lee and Chen, 2005] Lee, T. and Chen, N. (2005). A two-stage hybrid credit scoring model using artificial neural networks and multivariate adaptive regression splines. *Expert Systems with Applications*, 28(4):743–752.
- [Li et al., 2021] Li, G., Ma, H. D., Liu, R. Y., Shen, M. D., and Zhang, K. X. (2021). A two-stage hybrid default discriminant model based on deep forest. *Entropy*, 23(5):582.
- [Patel et al., 2020] Patel, B., Patil, H., Hembram, J., and Jaswal, S. (2020). Loan default forecasting using data mining. *Conference Proceedings*. Further details needed.
- [Rudin, 2019] Rudin, C. (2019). Stop explaining black-box machine learning models for high-stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215.
- [Sagming et al., 2024] Sagming, M., Heymann, R., and Visaya, M. V. (2024). *Using topological data analysis and machine learning to predict customer churn*. PhD thesis, University of Johannesburg.
- [Sheikh et al., 2020] Sheikh, M. A., Goel, A. K., and Kumar, T. (2020). An approach for prediction of loan approval using machine learning algorithm. *Journal Article*. Further journal details required.
- [Singh et al., 2007] Singh, G., Mémoli, F., and Carlsson, G. (2007). Topological methods for the analysis of high-dimensional data sets and 3d object recognition. In *Eurographics Symposium on Point-Based Graphics*.
- [Thomas, 2000] Thomas, L. C. (2000). A survey of credit and behavioural scoring: Forecasting financial risk of lending to consumers. *International Journal of Forecasting*, 16(2):149–172.
- [Warrior, 2015] Warrior, D. (2015). Tda 4: Persistence. <https://datawarrior.wordpress.com/2015/12/20/tda-4-persistence/>. Accessed: 2024-12-02.
- [Yadav and Singh, 2023] Yadav, K. and Singh, S. (2023). Loan status prediction using svm and logistic regression. *Conference Proceedings*. Further details needed.
- [Zomorodian, 2005] Zomorodian, A. (2005). *Topology for computing*. Cambridge University Press.
- [Zomorodian and Carlsson, 2005] Zomorodian, A. and Carlsson, G. (2005). Computing persistent homology. *Discrete & Computational Geometry*, 33(2):249–274.
- [Zulkepli et al., 2019] Zulkepli, N. F. S., Noorani, M. S. M., Razak, F. A., Ismail, M., and Alias, M. A. (2019). Topological characterization of haze episodes using persistent homology. *Aerosol and Air Quality Research*, 19:1614–1624.